TACEDGE · PRODUCT DEFINITION PACKAGE

TACEDGE Ground Engineering

V2.0 · Product Definition Package

The complete six-artefact product definition for the first Operational Workspace on the TACEDGE platform: a field-operations capture-and-trust system for frontline ground-engineering delivery. It carries the product from agreed intent through to the data architecture a technical owner builds against.

STATUS	CLASS	OWNER
V2.0 · revised edition	Internal	Product Owner (founders)

PRODUCT
TACEDGE Ground Engineering (Workspace B1)

ALIGNED TO
Brand Guidelines v1.1 · Configure · Capture · Confirm

THE DOCUMENT SET

Six artefacts, one derivation chain

The Product Brief fixes the problem, users, scope and the six-criterion Definition of Done. The Requirements Backlog sets prioritised scope with a clear MVP line. The Functional Architecture describes how the workspace behaves. The Page Definition Sheets define the screens. The Design System fixes the tokens and the discipline. The Technical & Data Architecture fixes the data, identity, state and interfaces beneath the surfaces.

ABOUT THIS EDITION

This revised edition brings the package into line with **Brand Guidelines v1.1** and the current build. Three changes run through every artefact: the operational spine is named **Configure · Capture · Confirm** (Release is the outcome of Confirm, not a fourth stage); the product is **TACEDGE Ground Engineering**, a six-work-type ground-engineering platform whose V2.0 release proves the spine in depth on anchoring and piling; and the visual system follows the v1.1 tokens and the single Play / Be Vietnam Pro / JetBrains Mono type system.

SHARED CLARITY, FROM THE OPERATOR TO THE ENGINEER

Executive summary

TACEDGE Ground Engineering is the first Operational Workspace (B1) on the TACEDGE platform: a field-operations capture-and-trust system for frontline ground-engineering delivery. This package is its complete product definition, six artefacts that carry the product from agreed intent through to the data architecture a technical owner builds against.

The problem is a broken information chain. Frontline work is still captured on paper, status is unknowable until someone compiles it, QA happens late, and the engineer waits for an end-of-job report before trusting anything. The operator, least incentivised and bearing the switching cost, is the make-or-break adopter: nothing downstream has value unless capture is instant and lossless in poor-connectivity field conditions. The commercial edge is an engineer-ready record trusted on sight, which wins and defends work and lowers risk at closeout.

V2.0 proves a configurable platform rather than a single product. One operational spine, **Configure, Capture, Confirm**, serves every work type; a work type is a completion-criteria configuration, not new code. Release is the outcome of Confirm, not a separate stage: the confirmed record is what reaches the engineer and the contract. The platform carries a catalogue of **six ground-engineering work types**, anchoring, piling and retaining, drilling, shotcrete, rockfall protection, and drainage; V2.0 proves the spine in depth on the discrete-element pattern, **anchoring and piling**, and delivers the full operator to PM to engineer chain: offline-first capture, a Status Board and a located Spatial Map, a QA queue that turns submitted work into a confirmed record, and a closeout package assembled from the live record. The remaining four types onboard as configuration of the same spine, not bespoke build.

THE PRODUCT MODEL

Configure. Capture. Confirm. The PM configures the job once; the operator captures ground truth with minimal decision load; the PM confirms it, and only the confirmed record is released outward. One work item carries one connected record from the hole to the engineer.

The six artefacts form a derivation chain. The Product Brief fixes the problem, users, scope and the six-criterion Definition of Done. The Requirements Backlog sets prioritised scope with a clear MVP line. The Functional Architecture describes how the workspace behaves as a system. The Page Definition Sheets define the screens. The Design System Spec fixes the tokens and the discipline. The Technical and Data Architecture fixes the data, identity, state and interfaces beneath the surfaces. A single locked decisions ledger keeps the set internally coherent.

Four architectural commitments run through the set. A work type is data, not code, so a new type is a configuration rather than a rebuild. Every record has a single owner at every review state, and only confirmed records reach the engineer or the contract: confirmed-only-outward. The confirmed record is expressible through the Operational Data Contract as five objects, the source-neutral seam the Coordination layer reads. The coordination-readiness seams, project geolocation, an organisation tier, first-class status and an addressable project, are kept reachable but not built in V2.0.

Status. The Phase 0 and Phase 1 gates are met and Phase 2 (Architecture) is open. The long-pole ahead is the Technical Owner appointment. The one assumption still carrying risk is the Operational Data Contract, which stays assumed until a second workspace or a real consumer exercises it.

FOUNDATIONAL GROUND-ENGINEERING FIELD PLATFORM · PHASE 0 FOUNDATIONS ARTEFACT

Product Brief

The first Operational Workspace on the TACEDGE platform. The apex of the document set and the source of truth for product intent; the Requirements Backlog derives from it.

STATUS	v3.2 · Foundations	CLASS	Internal
OWNER	Product Owner (founders)	PRODUCT	TACEDGE Ground Engineering (Workspace B1)
PARTNER	Groundline Civil (beachhead)	PHASE	0 · Foundations

· What this is

The Phase 0 Foundations artefact for TACEDGE Ground Engineering: the agreed problem, users, scope, Definition of Done and success metrics. It is the apex of the document set and the source of truth for product intent, and the Requirements Backlog derives from it.

v3.2: the Definition of Done is fixed as six numbered criteria, cited downstream as DoD 1 to 6; voice transcription is locked as a V2.0 Should.

01 Problem

Frontline ground-engineering work is still captured on paper: logbooks, safety checklists, sign-on sheets. Existing digital tools are too clunky and overbuilt for operators, so they go unused. The result is a broken information chain: manual, duplicated data entry; delay at every link; and no shared operational clarity for the PM or the engineer on progress, status and issues. And in poor-connectivity field conditions the current tools fall back to paper, breaking the chain exactly where the work happens.

And the cost is not only admin time. A slow, manual record means reporting reaches the engineer late and gets questioned; variations and claims rest on evidence that was never captured cleanly at source; and the contractor carries avoidable risk into closeout. Paper does not just slow the work, it weakens the contractor's commercial position.

Product problem. V1 has the capability but not the first-use UX. The operator, least incentivised and bearing the switching cost, is make-or-break: nothing downstream has value unless they adopt and the platform performs.

02 Users

ROLE	WHAT THEY DO	WHAT THEY GET
Operator (driller)	Captures work at the point of drilling. Frontline data entry.	Less paperwork, faster sign-off, more time on the tools.
Site supervisor	Holds a compliance and roster slice (not full PM visibility); often also drills. On site all day (tablet / phone), the integration test.	Compliance and roster handled in the flow of work, not after it.
Project manager	PM view. Often remote. Runs the project from the live feed; the single decision authority.	Live status without chasing; the project is known, not reassembled.
Engineer (client)	The commissioning party. Confirmed record only, with own permissions. Must trust the reporting on sight.	A record trusted on sight, delivered faster, with less back-and-forth.

Primary user. The operator. Adoption and reliability are the platform's load-bearing condition.

Economic buyer. The business owner. The engineer-ready report is the platform's sharpest commercial edge: stronger reporting that wins and defends work, clean evidence for variations and claims, lower risk at closeout, and a bidding advantage built on a record engineers trust on sight.

03 Scope, the MVP line

DIMENSION	IN V2.0	ROADMAP
Work types	Anchoring and piling, proven end-to-end on the discrete-element pattern. The shared catalogue also carries drilling, shotcrete, rockfall protection and drainage.	The remaining four catalogue types onboard as configuration on the same spine; further ground-engineering types beyond the six.
PM status view	Status Board (schematic, by zone) · Spatial Map (work items plotted on an uploaded site image)	Site Map (full geospatial: real space on image / LiDAR / drone)
Safety docs	JSA / SWMS uploaded, visible, acknowledged	In-product authoring
Project documents	Reference docs (drawings, site instructions, lift plans, engineer docs): uploaded, viewed, available offline. A project folder, not a DMS	Categories, search, links to work items
Incident & near-miss	Standalone capture in under a minute; PM alerted on submission	(none)
Daily activity	Resourcing sheet: crew and plant hours, auto-totalled, exported	Pay / invoice computation
AI assist	Plan import from PDF; test-record conversion, human-verified; voice transcription (Should)	Richer field extraction
Capability	Full operator → PM → engineer (client) chain, configurable per work type	Coordination layer

Ground engineering spans many work types; the platform scales by configuring a shared workflow, not by building a product per type. Every work type runs the same operational spine, **Configure, Capture, Confirm**, with only the completion criteria varying by type. V2.0 proves the spine on the discrete-element pattern (anchoring and piling); the other four catalogue types, and further types beyond them, onboard as configuration of that spine rather than bespoke build. The same operational spine carries the next products on the roadmap, including the Coordination Layer, on one operational model rather than a set of separate tools.

04 Definition of Done

1 Cold PM

A new PM logs in untrained, sets up and runs a project. "Holy sh*t, that was easy."

2 Operator

Replaces paper logbooks on the ground. "No more logbooks, this is so easy."

3 Operator → PM

Field capture populates the PM's live view. The PM sees the field, live.

4 Offline

The operator captures a full shift with no signal and loses nothing on resync. Paper is never the fallback.

5 PM → Engineer (client)

Reporting reaches the engineer (client) clean and trusted, no chasing. Trusted on sight.

6 Platform

Anchoring and piling run on one configurable spine; the config boundary is clean and documented, and the path to the full six-type catalogue is proven.

These six are the canonical Definition of Done. Downstream artefacts (Requirements Backlog, Functional Architecture, Page Definition Sheets, Technical Architecture) cite them by number, DoD 1 to 6. Extensibility across the six-type catalogue, and to further ground-engineering work types, is carried into Phase 2 as an architecture constraint, not a V2.0 gate.

05 Success metrics

TIER 1 · ACCEPTANCE, MEASURED AT LAUNCH

METRIC	TARGET
Crew sign-on (phone)	< 10 sec
Incident / near-miss submission	30–60 sec
PM setup (cold, client info only)	< 10 min
PM status on login	< 10 sec
PM job summary	0 re-entry

METRIC	TARGET
Paper displacement at rollout	100% / 2 wks

TIER 2 · COMMERCIAL, LAGGING, PROVES IT WORKED

→ Groundline Civil wins more work from engineers on the strength of reporting quality.

→ Engineers pull TACEDGE to other contractors as the expected reporting standard.

→ Variations and claims supported by source-captured evidence.

→ Closeout packaged from the live record, not reconstructed.

→ Reporting quality cited in bid wins.

06 Phase 0 stage gate

Are the problem, the users, and the definition of success written down and agreed?

GATE VERDICT · YES

Problem, users, scope, definition of done and metrics are written down and agreed. Phase 1 (Discovery & Requirements) is complete and Phase 2 (Architecture) is open.

PHASE 1 DISCOVERY & REQUIREMENTS ARTEFACT · REQUIREMENTS COMPLETE · OPEN DECISIONS RESOLVED · PHASE 2 ARCHITECTURAL CONDITIONS

Requirements Backlog

Prioritised V2.0 scope with a clear MVP line, derived from the Product Brief. The full operator-to-client chain specified across the ground-engineering catalogue, with open decisions resolved and the Phase 2 architectural conditions identified.

STATUS	v1.0 · Baseline	CLASS	Internal
OWNER	Product Owner (founders)	DERIVES FROM	Product Brief V2.0 (v3.2)
PARTNER	Groundline Civil (beachhead)	PRODUCT	TACEDGE Ground Engineering (Workspace B1)

01 Where this sits

Requirements capture for V2.0 is functionally complete, and the open decisions are now resolved. The full operator-to-client chain is specified for both work types in scope, anchoring and piling (the shared six-type catalogue also carries drilling, shotcrete, rockfall protection and drainage), and piling validates the platform bet: it needed no new workflows, only a configuration. Records & Surveys (V1 tabs) have been reviewed and consciously deferred from V2.0. The second gate clause, the riskiest-assumption checks held in Open decisions, has been worked through with Groundline Civil and a real driller. The Phase 1 gate is therefore met; Phase 2 (Architecture) opens, with the Technical Owner appointment as its long-pole prerequisite.

02 MVP scope (from Phase 0)

In V2.0: drill logs for anchoring and piling, the V2.0 proof pair on the discrete-element pattern, while the shared catalogue also carries drilling, shotcrete, rockfall protection and drainage · full operator-to-client chain, configurable per work type · Status Board (grid, clickable, by zone) · spatial map (anchors plotted on an uploaded site image) · JSA / SWMS uploaded, visible, acknowledged.

Roadmap: the four remaining catalogue types, drilling, shotcrete, rockfall protection and drainage, onboarding as configuration on the same spine · Site Map (full geospatial) · coordination layer · in-product safety-doc authoring · crew briefings and toolbox talks.

03 Work Item status model

Each work item carries two lifecycle dimensions that move through states, work phase and review, with two overlays riding alongside them, the testing flag and freshness. A work item carries one value from each at once, so the Status Board and the QA queue can tell the truth simultaneously. Conflating any of them into one field would make that impossible.

TWO LIFECYCLE DIMENSIONS · TWO OVERLAYS

DIMENSION	VALUES	DRIVES
Work phase	Planned · Active · Complete, plus off-path Requires-redrill and Not-required. The anchor read is Drilled · Grouted · Tested.	Status Board tallies and colours. Auto-advances from operator actions; PM can override.
Review state	In progress · Submitted for QA · Confirmed · (Re-opened)	Editability and engineer visibility (the log lifecycle below).
Testing flag · overlay	Flagged / not flagged for load testing	Independent boolean; not every anchor is tested.
Freshness · overlay	Fresh · Ageing · Stale	How current the shown status is, derived from sync state; an unrefreshed status reads as ageing, not settled fact.

EXAMPLE

An anchor can be Grouted (phase), still In progress (review), and flagged for testing, all at once. Requires-redrill is one event with two effects: it sets the work phase and fires the PM alert (ALT-1).

ANCHOR LOG LIFECYCLE (REVIEW STATE)

STATE	WHO CAN EDIT	ENGINEER SEES
In progress	Operator (Save at any stage)	No
Submitted for QA	PM only (operator locked out)	No
Confirmed	No one	Yes
Re-opened	PM only (reverts to QA)	No, until re-confirmed

04 Design principles surfaced

These principles are carried to Phase 3. They are the reason the requirements take the shape they do.

→ **Operator defaults to confirming and recording, not deciding.** The PM front-loads every decision into setup; the operator touches only what physically varies and what they observe.

→ **Exceptions are operator-declared, not system-inferred.** Redrill and test-fail (anchor-bound) and incident/near-miss (standalone) are a single operator action that alerts the PM, never inferred from data.

→ **AI proposes, human approves, then commits.** PDF plan import proposes zones and counts; testing conversion proposes extracted data; the PM/operator approves before anything is committed or trusted.

→ **For testing, value is in digitising the output, not the capture.** Paper wins at the gauge; the platform adds value downstream via AI conversion.

→ **Status auto-advances from the field, with PM override.** Operator actions move the work phase live; the PM can correct or set off-path states by hand.

→ **Records are independent, not gated chains.** The natural daily sequence is encouraged, not enforced; gating is reserved for acts an individual can always complete alone.

→ **Acknowledgement is individual and attributable.** Sign-on signatures are made through the person's own authenticated session, never applied in bulk by a supervisor.

→ **Gate only on acts an individual can always complete alone.** JSA sign-on qualifies and gates drilling; nothing that needs another person ever gates.

- → **Editability and visibility are single-owner at every review state.** Removes the risk of a client deliverable changing silently.
- → **Changes to money- or client-facing data must be visible.** Wherever a record drives payment or reaches a client, edits carry a light who-changed-what-when trail (QA-3, DAS-6).
- → **PM views are three layers: inform, act, release.** Overview informs, Drill Summary acts, Reporting releases. The same exception is surfaced, then resolved, then released, never one screen.
- → **Calculate the measurable variances; declare the judgement calls.** Grout/depth variance is system-computed against design; safety/work exceptions stay operator-declared.
- → **The system surfaces every finding, variance or exception; the PM decides which reach the client.** Surface everything internally; release selectively. Discretion for clarity, not concealment.
- → **The QA queue is a two-way channel; re-open is the universal hand-back.** Submit-for-QA means done or needs-your-eyes; re-open hands an anchor back whether confirmed or submitted early. One mechanism, not several.
- → **The engineer view is glance-first.** Every element earns its place by being readable in seconds; detail is one click away, but the state of the job must be immediate.
- → **The PM curates client exposure by the source of an issue.** Ground reality tends toward the client; contractor performance is reviewed and resolved first. For clarity and fairness, not concealment.
- → **A work type is a completion-criteria configuration on one spine.** Every work type runs the same operational spine: Configure, Capture, Confirm. Drill, grout and test are the Capture configuration of the discrete-element pattern; the work type defines which stages are required for a valid submission. A new work type is a moved done line, not new capability.
- → **Set the conditions, don't build the feature.** Where a roadmap capability depends on data-model shape, Phase 2 must keep it reachable, a config or a zoom away, not a rebuild away, without building it in V2.0.

05 User stories

Every requirement below is a V2.0 Must; priority within V2.0 is build order, not dropped items. The criterion-critical Musts, those that prove the Brief's six Definition-of-Done criteria (DoD 1 to 6), are the first build wave: setup and plan (DoD 1, Cold PM), operator capture (DoD 2, Operator), the live operator-to-PM view (DoD 3, Operator to PM), offline-resilient capture (DoD 4, Offline), QA to a trusted client record (DoD 5, PM to

Engineer), and work-type configuration (DoD 6, Platform). Test capture and verification (TST) is the evidence path that proves the trusted record, not a criterion of its own.

PROJECT SETUP

The invariant project shell: project, zones and safety docs. Work-type-agnostic.

PM PROJECT SETUP · PROJECT & STRUCTURE

The building blocks are assembled into Work Item Templates; the Work Plan places templates as planned work items by zone. Every preset here is one less thing the operator touches, and this is where the sub-10-minute setup metric is won.

ID	REQUIREMENT	PRIORITY
SET-1	**Create a new project.** Container for specifications, plan and logs; carries a project code.	MUST
SET-2	**Define zones.** Zones (Zone 1/2/3, add more) group the anchor plan and board. Each zone has a row-direction setting (LTR/RTL).	MUST
SET-3	**Upload JSA/SWMS.** PM uploads the project JSA and SWMS at setup and replaces them when new versions become relevant. The consumption side is the operator sign-on (SOS-1–3).	MUST

OPERATIONAL SETUP

The work-type-configured layer (PLAT-3): specifications, templates and the plan are parameterised per work type.

SPECIFICATIONS · BUILDING BLOCKS

ID	REQUIREMENT	PRIORITY
SPEC-1	**Define drill methods.** Named methods (e.g. Down-the-Hole, Top Hammer – Air Flush) for controlled operator selection.	MUST
SPEC-2	**Define grout mix types.** Base categories (e.g. Epoxy, Microsilica), the kind of grout.	MUST
SPEC-3	**Define grout mixes.** Named recipes with strength spec (MPa), built on a type. The litres-per-bag / volume logic the operator inherits (GRT-2) lives here.	MUST
SPEC-4	**Define lithology options.** Project material list operators select from when logging by depth (DRL-3).	MUST

WORK ITEM TEMPLATES · ASSEMBLED FROM BLOCKS (PROJECT-LEVEL, ALL ZONES)

ID	REQUIREMENT	PRIORITY
DSN-1	**Define work item template.** Reusable spec: Template Name, Anchor/Bar Type, Bar Spec, Hole Diameter, Max Depth, Bond/Debonded Length, Drill Angle, Drill Method, Grout Mix, Design Load (kN, optional), Notes. Required: Name, Bar Type, Bar Spec, Hole Diameter, Max Depth.	MUST
DSN-2	**Carry-over / start-blank.** A new template pre-fills from an existing one (edit what's different). The key setup accelerator; most templates are variations.	MUST
DSN-3	**Inline create block.** + New beside Drill Method and Grout Mix creates a block without leaving the template form, no setup dead-end.	MUST

WORK PLAN · ZONES TO ROWS TO WORK ITEMS

ID	REQUIREMENT	PRIORITY
PLAN-1	**Plan anchors by row.** PM defines rows (letter prefix + count + template); anchors auto-generate and auto-save. Drillers see saved anchors in Drill Logs immediately.	MUST
PLAN-2	**Import plan from PDF.** Upload the design drawing + plain-language scope rules; AI proposes zones and counts; PM reviews and approves before any rows are created. Major lever on the 10-minute setup metric.	MUST
PLAN-3	**Manage a planned anchor.** Open any anchor to set work-phase status, reassign template, flag/unflag for testing, add notes, or delete, correcting the plan after generation.	MUST

SIGN-ON & SAFETY · SIGN-ON (OPERATOR)

Greenfield in V2.0. The operator's day starts with JSA sign-on, the one act that gates work.

ID	REQUIREMENT	PRIORITY
SOS-1	**Daily sign-on.** Operator acknowledges the JSA and SWMS with one tap and a finger signature, date auto-captured: a deliberate, attributable compliance record in seconds.	MUST

ID	REQUIREMENT	PRIORITY
SOS-2	**Documents available to reference.** Current JSA and SWMS are available to open if the operator chooses, without being forced to read them to sign on.	MUST
SOS-3	**New-document soft flag.** A new or updated JSA/SWMS is flagged to the operator so they can see it changed; it does not block or alter the sign-on workflow.	MUST
SOS-4	**Sign-on gates drilling.** System requires that day's JSA sign-on before the operator can begin a drill log, the one enforced field barrier, always satisfiable regardless of crew size.	MUST

PROJECT DOCUMENTS · REFERENCE LIBRARY

A lightweight project folder for inbound reference material, added at SteepWorks pilot input: a place to keep the drawings and instructions a job runs on, distinct from safety sign-on (SOS) and spatial context (SPC), and deliberately a folder rather than a document-management system.

ID	REQUIREMENT	PRIORITY
DOC-1	**Project reference documents.** A project carries a documents area where reference material (drawings, site instructions, lift plans, engineer-issued documentation and other project reference) can be uploaded, categorised and viewed, distinct from JSA / SWMS which keep their sign-on gate (SOS).	SHOULD
DOC-2	**Offline availability.** Project reference documents are available to the operator offline, alongside other inherited project content.	SHOULD
DOC-3	**Project folder, not a DMS.** No version control, approval workflow or in-app editing; the area reuses the evidence / file model and is read-and-reference only. Acknowledgement and the sign-on gate stay with SOS.	SHOULD

COMPLIANCE VISIBILITY

ID	REQUIREMENT	PRIORITY
SAF-1	**Outstanding sign-on visibility.** Supervisor/PM can see who is logged onto the project but hasn't completed that day's JSA sign-on: a live compliance picture.	MUST

OPERATOR DRILL-LOG SLICE · DRILLING

ID	REQUIREMENT	PRIORITY
DRL-1	**Select the planned anchor.** Operator selects the pre-planned anchor they are drilling from Drill Logs; its template and inherited specs are already attached. Gated on that day's JSA sign-on (SOS-4).	MUST
DRL-2	**Confirm presets & start.** Drilling parameters are preset by the PM; angle is shown as informative guidance, not an input. Operator confirms and starts: minimal selection, reduced decision load.	MUST
DRL-3	**Log lithology by depth.** Operator records depth intervals (from-to) and selects material from the SPEC-4 preset list, adding layers to total depth. Schema (from/to + material) confirmed with a real driller and locked.	MUST
DRL-4	**Add notes.** Free notes by text: water strikes, voids and anything unusual. Voice transcription is a Should, not a Must.	MUST
DRL-5	**Auto-capture drilling date.** System captures the drilling date automatically (display-and-confirm). Override permitted for offline-sync, back-fill and next-morning cases. Full timestamp stored; the log displays the date only.	MUST
DRL-6	**Mark as drilled.** Operator closes the drilling phase; anchor work phase auto-advances to Drilled.	MUST

GROUTING

ID	REQUIREMENT	PRIORITY
GRT-1	**Select grout & auto-date.** Operator selects grout for a drilled anchor; grout date auto-captured, independent of the drill date.	MUST
GRT-2	**Cement & auto-volume.** Operator enters number of bags and selects bag size (20/25/40 kg); system auto-derives grout volume from the mix's preset litres-per-bag factor (SPEC-3).	MUST
GRT-3	**Select grout mix.** Operator selects mix from PM presets.	MUST
GRT-4	**Record additives.** Operator adds additives (e.g. Blue Chem) and bag counts.	MUST
GRT-5	**Grout return achieved.** Tick-box QA indicator captured at point of work.	MUST

ID	REQUIREMENT	PRIORITY
GRT-6	**Grout sample taken.** Records whether a sample was taken (7/28-day). V2.0 records the fact only; results-return is roadmap.	MUST
GRT-7	**Requires redrill flag.** Operator flags grout loss/failure. One event, two effects: sets work phase to Requires-redrill and fires PM alert (ALT-1).	MUST
GRT-8	**Grouting notes.** Free notes by text: partial return, void encountered, pressure issues. Voice transcription is a Should, not a Must.	MUST
GRT-9	**Mark as grouted.** Operator closes the grouting phase; work phase auto-advances to Grouted.	MUST

ANCHOR TESTING

ID	REQUIREMENT	PRIORITY
TST-1	**Photo-capture test record.** Operator photographs the written test record and submits it, keeping the fast paper capture they trust.	MUST
TST-2	**AI conversion.** System converts the photo into the configured, report-ready testing format. Core value.	MUST
TST-3	**Verify against photo.** PM/operator reviews AI-extracted data alongside the original photo and confirms or corrects. Trust gate for the confirmed record the engineer trusts on sight (DoD 5).	MUST
TST-4	**Test-failed flag.** Operator declares a failed test with a button at submission (not inferred from the photo); fires PM alert (ALT-2).	MUST

CLOSE-OUT

ID	REQUIREMENT	PRIORITY
CLO-1	**Save in progress.** Operator can Save at any stage before submission: capture progressively without submitting prematurely.	MUST
CLO-2	**Submit for QA (any phase).** Single clear action hands the anchor to the PM and locks operator editing. Can be submitted at any work phase, enabling early escalation.	MUST
CLO-3	**Confirm-before-submit.** A simple confirmation prompt prevents accidental pre-emptive submission.	MUST

PM · QA, ALERTS & LIFECYCLE

ID	REQUIREMENT	PRIORITY
QA-1	**Anchor into QA queue.** Submitted anchor appears in the PM queue with its compiled log (drilling, grouting, testing) ready to review: zero manual re-entry.	MUST
QA-2	**Edit during QA & Confirm.** PM edits during QA, then Confirms; on confirmation the anchor finalises and becomes visible to the engineer.	MUST
QA-3	**Re-open / hand back.** PM-only. Hands an anchor back to the operator, whether confirmed or submitted early, releasing the edit lock and dropping it from client view until re-submitted/re-confirmed. A light who-and-when record is kept (V2.0).	MUST
ALT-1	**Redrill alert.** Redrill flag (GRT-7) surfaces as an alert against the anchor in the PM live view.	MUST
ALT-2	**Test-fail alert.** Test-failed flag (TST-4) surfaces as an alert against the anchor in the PM live view.	MUST
ENG-1	**Engineer sees confirmed only.** Engineer view shows only Confirmed anchors, never in-progress or unreviewed data.	MUST

INCIDENT & NEAR-MISS

Standalone, not anchor-bound; an incident or near-miss can happen anywhere on site, anytime, with its own entry point. Built for the 30–60-second target.

ID	REQUIREMENT	PRIORITY
INC-1	**Capture an incident/near-miss.** Operator reports an incident or near-miss with a few required fields (type; injury y/n; severity Low/Med/High) plus a typed narrative, and optional photos/video, well under a minute. Severity scale confirmed against Groundline Civil's scale; voice a Should.	MUST
INC-2	**Required criticals protect omission.** The few genuinely critical facts are required structured input, so an operator under pressure cannot leave out something the PM must know.	MUST
INC-3	**Triage-ready PM alert.** Submission alerts the PM immediately, with type, injury flag and severity visible in the alert itself (project-level).	MUST

DAILY ACTIVITY SCHEDULE

The supervisor's resourcing record: the one part of the field domain about cost rather than the work itself. It feeds payroll (primary) and invoicing (a PM option). V2.0 captures and exports the hours; pay/invoice computation is roadmap.

ID	REQUIREMENT	PRIORITY
DAS-1	**Daily activity sheet.** Supervisor submits one sheet per project per day, date defaulted to today but editable. Submit-and-visible to the PM, no QA/approval gate.	MUST
DAS-2	**Staff roster pre-filled.** Today's signed-on crew is pre-listed with names and roles (SOS-1); anyone not in the system is added manually.	MUST
DAS-3	**Staff time & auto-total.** Per person: Start, Finish, Travel, Break; Total hours auto-calculated. Supervisor does no arithmetic.	MUST
DAS-4	**Bulk-fill, edit by exception.** Supervisor fills one row of times and propagates to all staff, then changes only the exceptions.	MUST
DAS-5	**Plant library.** PM defines the project's plant/equipment in setup; supervisor can add plant on site. Plant time recorded as working hours from presets.	MUST
DAS-6	**Post-submission edit record.** Sheet editable after submission by supervisor or PM, with a light who-changed-what-when record.	MUST

PM VIEWS · OVERVIEW (INFORM)

The PM's side of the chain is three layers: Overview informs, Drill Summary acts, Reporting releases.

ID	REQUIREMENT	PRIORITY
PMV-1	**Project pulse at a glance.** PM sees status within seconds of opening: progress by work phase, by zone, and anything needing attention, the sub-10-second status metric.	MUST
PMV-2	**Live drill-log status.** A constant, live view of drill-log progress and what's awaiting QA.	MUST
PMV-3	**Operational alerts surfaced.** Surfaces test-completed, redrill/collapse (ALT-1), incident/near-miss (INC-3, ALT-2), activity-sheet submissions as they happen.	MUST

ID	REQUIREMENT	PRIORITY
PMV-4	**Grout variance flag.** System flags grout volume excess/shortfall against expected, calculated automatically.	MUST
PMV-5	**Depth variance flag.** System flags drilled depth excess/shortfall against the design depth automatically.	MUST
PMV-6	**Review safety events.** Logged incidents and near-misses are reviewable, not just alerted-and-gone.	MUST

DRILL SUMMARY (ACT)

ID	REQUIREMENT	PRIORITY
DSM-1	**Work item table.** Every anchor as a row: Ref, location/zone, date/time, length, diameter, crew, work-phase status, test state, review state.	MUST
DSM-2	**Filter & search.** Filter by zone, status and time; search by reference/crew/comments.	MUST
DSM-3	**Review entry point.** A Review action on submitted anchors opens the log directly from the table to start the confirm cycle (QA-1/2/3).	MUST
DSM-4	**Compiled review detail.** Opening an anchor shows the whole compiled record, drilling, grouting and testing, with variance flags and the original test photo.	MUST
DSM-5	**Variances & exceptions as filters.** System-derived variances and operator-declared exceptions appear as filterable indicators in the same table, no separate issues screen.	MUST

REPORTING (RELEASE)

ID	REQUIREMENT	PRIORITY
RPT-1	**Whole-project export.** Export the entire project's confirmed record as the project closeout package: read-only PDF and CSV, assembled from the live record rather than reconstructed at end of job. A coherent, trusted deliverable for the engineer, zero manual re-entry.	MUST
RPT-2	**PM-controlled exception annotation.** PM chooses whether to export with exceptions annotated, or to remediate them first.	MUST

ID	REQUIREMENT	PRIORITY
RPT-3	**Push progress to client view.** PM makes a progress summary viewable to the engineer (client) (ENG-1), optionally with selected concerns annotated at the team's discretion.	MUST
RPT-4	**Variation and claim evidence pack.** A filtered export assembling variance flags (PMV-4/5), operator-declared exceptions, the who-changed-what trails (QA-3, DAS-6) and supporting photos into the contractor's evidence base for variations and claims. Assembles data already captured.	SHOULD
RPT-5	**Report format by work type.** The export format is a work-type configuration (PLAT-3), so each work type releases in its configured, report-ready shape (TST-2).	MUST

ENGINEER (CLIENT) VIEW

The far end of the chain: the engineer (client) sees but never captures or edits. Governed throughout by ENG-1 (confirmed-only). Its achievement is timing: the record builds in near-real-time instead of waiting for an end-of-job report.

ID	REQUIREMENT	PRIORITY
ENG-2	**Live progress board, glance-legible.** Engineer sees the Status Board with full planned scope and progress overlaid, each anchor's status clear at a glance (colour/state) without clicking.	MUST
ENG-3	**Progressive confirmed report.** Engineer drills into the full confirmed record, lithology, grout, anchor testing results, for every anchor confirmed and made viewable, delivered as work is QA'd.	MUST
ENG-4	**Exceptions PM-gated by source.** Exception states are withheld from the engineer until the PM reviews and chooses to surface them: ground-driven issues often shared, performance-driven faults resolved first.	MUST

PILING · THE CONFIGURABILITY PROOF

Piling is the second work type, and it proves the platform bet. It is drilling, minus grouting and testing: same drilling, same template presets, same lithology-by-depth. So piling needs no new workflows; it is a configuration of the existing spine.

ID	REQUIREMENT	PRIORITY
PIL-1	**Work type = required-phase configuration.** A work type defines which phases are required for a valid submission: anchors require drill+grout+test; piling requires only drill. Proves DoD 6 (Platform).	MUST

ID	REQUIREMENT	PRIORITY
PIL-2	**Piling drilling = anchor drilling.** Identical drilling workflow, template presets, lithology by depth, auto-date, notes, submit-for-QA. Nothing re-learned or rebuilt.	MUST
PIL-3	**Drill-only submission is valid.** A piling log is complete and submittable at drilling; grout and test phases are available if scope includes them, but never block submission. Flows through QA, reporting and the engineer view identically.	MUST

PLATFORM REQUIREMENTS · WORK ITEM, SPINE AND CONFIGURATION

These make the platform explicit. They add no new build in V2.0 (still only the two configured work types) but they name the work-item model, the configuration object and the fixed spine so the Technical Owner commits the right data model. The capture tables above are the anchor and shared configuration; they operate on this generic model.

ID	REQUIREMENT	PRIORITY
PLAT-1	**Work item model.** A located, installable element with identity (PLAT-9), location, template, status and log. Anchor and pile are types of work item; Template, Plan and Board operate on work items, not anchors.	MUST
PLAT-2	**Work type declaration and selection.** A project declares its work type(s); every work item carries its type. V2.0 configures two: anchor and pile.	MUST
PLAT-3	**Work type configuration object.** A work type defines its required stages, fields, validations, report format and completion criteria as data, not code.	MUST
PLAT-4	**Operational spine is fixed.** Configure, Capture, Confirm is invariant across work types; only the content of each stage varies by configuration. Release is the outcome of Confirm, the confirmed record released outward, never a fourth stage.	MUST
PLAT-5	**Generic completion criteria.** A valid submission is the work type's required stages and fields (anchors: drill, grout, test; piling: drill). Generalises PIL-1 and proves DoD 6 (Platform) at platform level.	MUST
PLAT-6	**Generic status surface.** The Status Board renders any work-item type by phase and zone; the anchor view is one rendering of it.	MUST

ID	REQUIREMENT	PRIORITY
PLAT-7	**Project status is derived and first-class.** A project carries a status derived from its work items: tallies by work phase, overall percent complete, open exception and alert counts, QA backlog, last-activity time, and a single coarse state token (Not started / Active / Blocked / Complete). Computed and stored as a queryable property, refreshed as work items change, and exposed through the ODC (ODC-1). Per-project derivation is V2.0 (it underpins PMV-1); cross-project aggregation is the ODC seam, not built. Satisfies ARCH-3.	MUST
PLAT-8	**Findings: variance, exception, alert.** Two finding types. A variance is system-computed against design (depth PMV-5, grout PMV-4). An exception is operator-declared judgement (redrill GRT-7, test-fail TST-4, incident/near-miss INC). An alert is the surfacing of either to the PM (ALT-1/2, INC-3), not a third type. Filtering, reporting and AI key off the finding type. No new screen; variances and exceptions stay inline (DSM-5).	MUST
PLAT-9	**Work item identity: system ID and human reference.** Each work item carries (a) an immutable system identifier, unique platform-wide, used by the ODC, Coordination and any external reference; and (b) a human-readable reference (e.g. B12), unique within its project or zone, shown to users and re-labellable without changing the system ID. Elaborates PLAT-1.	MUST
PLAT-10	**Evidence is a platform concept.** An evidence item is a captured artefact (photo, video, file) with provenance: capturer, timestamp and source (camera or upload), attached to a parent entity (work item, stage record, incident, daily sheet). Existing capture points (TST-1, INC-1, SPM-1) are instances; RPT-4 is a query over evidence. Models what is already captured; no media library and no new capture workflow.	MUST
PLAT-11	**Project lifecycle state.** A project carries an authored lifecycle flag (Open / Closed), PM-set at closeout and distinct from the derived status rollup (PLAT-7). Closed is declared by the PM, never computed; it withdraws the project from active delivery views and travels on the ODC. The derived coarse-state tokens (PLAT-7) are unchanged.	MUST

SPATIAL MAP · LOCATE ANCHORS ON SITE (V2.0 MUST)

The on-site identification surface, made Must in V2.0 after Groundline Civil discovery. A plain Status Board cannot tell a driller whether they are standing at B12 or B13; a plotted image can. Operator-facing and load-bearing for adoption; the same plotted map can also surface to the PM and engineer. This is manual plotting on an uploaded image; full georeferenced placement remains the Site Map roadmap item.

ID	REQUIREMENT	PRIORITY
SPM-1	**Upload a site base image.** PM uploads a base image for the project or zone: the engineer drawing, a site photo, or a drone or aerial image. This is the backdrop anchors are plotted on.	MUST
SPM-2	**Plot anchors on the image.** PM places each planned anchor as a pin at its real location on the base image, by anchor ID. AI-assisted placement from the engineer drawing where the plan was imported (PLAN-2); manual placement otherwise.	MUST
SPM-3	**Operator spatial view, status-coloured.** On site the operator sees the base image with every anchor pinned and coloured by work phase, alongside the Status Board, so they can identify exactly which anchor ID they are about to drill (is this B12 or B13?). Resolves the core on-site identification problem.	MUST
SPM-4	**Tap-to-select for drilling.** Tapping an anchor pin selects that planned anchor for its drill log (DRL-1), removing ID ambiguity at the point of work.	MUST
SPM-5	**Anchor carries an image position.** Each anchor persists an image-relative position (x/y on the base image) with the plan. A data-model field that is cheap now and expensive to backfill; the Technical Owner carries it from day one.	MUST

06 Resolved decisions

All Open decisions are now resolved, with Groundline Civil and a real driller where external input was required. None remain outstanding.

ITEM	RESOLUTION
Lithology schema (DRL-3)	Confirmed. from/to + material is the right model, validated with a real driller. Locked.
Auto-date override (DRL-5)	Yes. Override permitted for offline-sync, back-fill and next-morning cases.
Date vs timestamp	Store the full timestamp under the hood; the drill log displays the date only.
Voice transcription (DRL-4/GRT-8/INC-1)	Should, not Must. Typed text is the Must baseline; voice rides along as a Should.

ITEM	RESOLUTION
Light re-open record (QA-3)	In V2.0. Record who and when on re-open / re-confirm of a confirmed anchor.
Severity scale (INC-1)	Low/Med/High confirmed against Groundline Civil's safety-reporting scale.

07 Roadmap & feasibility flags (parked)

TECHNICAL OWNER FEASIBILITY FLAGS

→ **Offline reliability & sync:** the entire operator thesis depends on rock-solid capture in poor-connectivity field environments (incident photo/video especially; capture must succeed on-device instantly and sync later).

→ **Voice transcription offline in noisy field conditions.** A V2.0 Should; typed text is the Must baseline, so the offline-noise risk does not block the build.

→ **AI conversion accuracy** on handwritten field test sheets (TST-3 verify step de-risks).

→ **AI plan extraction** from PDF drawings (PLAN-2 approve step de-risks).

ROADMAP BANKED

→ **Crew briefings and toolbox talks.** Supervisor-recorded, with individual self-confirm attendance through each person's own login. Parked from V2.0 at Groundline Civil's direction; promote when a customer needs the record.

→ **Grout cost derivation** · sample-results return (7/28-day) · native digital test entry (only if faster than paper) · richer structured lithology capture.

→ **Site Map (full geospatial):** true georeferenced or LiDAR placement. The V2.0 spatial map already covers manual plotting on an uploaded image; this unlocks automated geospatial capture.

→ **Pay & invoice computation** (rates × hours); V2.0 captures and exports hours.

→ **Multi-contractor interoperability:** cross-organisation shared records. Sits with the coordination layer.

08 Architectural conditions for Phase 2

Two roadmap capabilities depend on data-model shape decided in Phase 2, not on V2.0 features. Neither is built in V2.0; both must remain reachable. These are day-one knowledge for the Technical Owner, because they shape the data model, the one thing that cannot be cheaply changed later.

1 · CONFIGURABILITY · NEW WORK TYPES (DOD 6 · PLATFORM)

The work-type-as-required-phase configuration (PIL-1) must be a true configuration boundary: a structurally different civil-construction work type (shotcrete, rockfall protection or drainage, not a drill-log) addable without re-architecting. Anchoring and piling prove the mechanism in V2.0; the architecture must remain fit for a type that isn't a drill log at all.

2 · THE COORDINATION LAYER · GEOSPATIAL COMMON OPERATING PICTURE

The strategic apex of the thesis: an organisation-wide map (e.g. a Fulton Hogan COO sees every live project as a pin on a map of NZ, clicks one for a status popover, descends into the project workspace). It is not new data; it is each workspace's confirmed record, expressed through the Operational Data Contract and aggregated above it. Required architectural shapes, even though no common operating picture is built in V2.0:

- Every project carries a geospatial location (lat/long), not just a name: cheap to add now, expensive to backfill.
- An organisation/tenant tier above projects, so the model knows an org owns many projects and status can aggregate by org.
- Project status is a first-class, queryable, aggregatable property exposed upward, not computed by opening the project.
- The project workspace is addressable and enterable from outside itself; click-pin-descend is a first-class navigational path.
- The confirmed record is expressible as the Operational Data Contract (status, progress, incident, risk): the source-neutral seam Coordination and multi-contractor interoperability both read.

ARCHITECTURAL REQUIREMENTS · COORDINATION READINESS

The Phase-2 architectural conditions above, promoted to identified requirements because the Technical Owner commits the data model from this backlog. Architect for these; build none of them in V2.0.

ID	REQUIREMENT	PRIORITY
ODC-1	**Operational Data Contract.** The confirmed record is expressible in a stable, source-neutral shape carrying status, progress, incidents, risk and location: the seam Coordination and multi-contractor interoperability both read. No consumer built in V2.0.	COND
ARCH-1	**Project geolocation.** Every project carries a lat/long, not just a name. Cheap to add now, expensive to backfill.	COND

ID	REQUIREMENT	PRIORITY
ARCH-2	**Organisation tier above projects.** An org/tenant tier, so the model knows an organisation owns many projects and status can aggregate by org.	COND
ARCH-3	**Status is first-class and aggregatable.** Project status is a queryable, aggregatable property exposed upward, not computed by opening the project (defined by PLAT-7).	COND
ARCH-4	**Workspace is externally addressable.** The project workspace is addressable and enterable from outside itself; click-pin-descend is a first-class navigational path.	COND

THE STANDING RULE

Build neither in V2.0. Architect so the coordination layer reads an addressable, status-emitting project through the Operational Data Contract, and a new work type is a configuration, not a rebuild.

09 Phase 1 gate · met

Requirements capture is functionally complete, Records/Surveys are consciously deferred, and the open decisions are resolved. The Phase 1 stage gate has two clauses, and both are now answered: prioritised backlog with a clear MVP line, and the riskiest-assumption checks worked through with Groundline Civil and a real driller.

What now stands in front of Phase 2 (Architecture) is not a capture task but a resourcing one: the Technical Owner appointment, the long-pole role the V1 effort lacked, required before Phase 2 can start. A light confirmation pass with unbiased Groundline Civil users remains good practice alongside early Phase 2 work, but it no longer blocks the gate.

PHASE 1 STAGE GATE · MET

Is there a prioritised backlog with a clear MVP line, and are the riskiest assumptions checked? Backlog + MVP line complete · riskiest-assumption checks resolved with Groundline Civil and a real driller. Phase 2 opens; Technical Owner is the long-pole.

FIRST OPERATIONAL WORKSPACE (B1) · PHASE 2 ARCHITECTURE ARTEFACT

Functional Architecture

How the workspace operates as an integrated system, the layer between approved requirements and implementation.

STATUS	v1.6 · Draft for review	CLASS	Internal
OWNER	Product Owner (founders)	AUDIENCE	Technical Owner · UX · Developers · PO
DERIVES FROM	Requirements Backlog (v0.17, B1)	PRODUCT	TACEDGE Ground Engineering (Workspace B1)

HOW TO READ THIS DOCUMENT

This document describes how the workspace behaves as an operating system: how it is organised, how information moves, how records change ownership and state, and which boundaries must not be broken. It is one level more abstract than the Requirements Backlog (what the product must do) and one level less abstract than the Product Brief (why it exists). It is not a backlog, a feature list, or a technical design. The Requirements Backlog V2.0 remains the source of truth for scope; where the two differ on what is in scope, the backlog wins.

01 Purpose

THE OPERATIONAL PROBLEM

Ground-engineering work runs as a chain of parties, operator, supervisor, project manager, engineer, client, but the record that joins them is fragmented. Field work is captured on paper, status is unknowable until someone compiles it, quality assurance happens late, and the client waits for an end-of-job report before they can trust anything. The result is operational fog: no party can see the true state of the job at the same moment, and problems surface too late to act on cleanly.

PURPOSE OF THE WORKSPACE

TACEDGE Ground Engineering carries a single, trusted record of what happened in the ground from the operator who drilled it to the engineer and client who must rely on it. It makes the operator-to-client chain continuous and legible: the operator captures ground truth with minimal decision load; the PM reviews, corrects and confirms it; the engineer and client watch a trusted record build in near-real-time rather than waiting for the end. Every record has one owner at every moment, every exception is visible internally, and only confirmed data is ever exposed outward.

POSITION WITHIN THE TACEDGE PORTFOLIO

This is the first Operational Workspace (B1) in the TACEDGE portfolio, a domain workspace built on the TACEDGE Shared Workspace Platform, and the capture end of the system. It produces ground truth and emits it, once confirmed, as a source-neutral summary onto the Operational Data Contract. TACEDGE Coordination (Product A) reads that contract to build its operating picture; Coordination is a separate product tier, not a layer of this workspace, and not a platform this workspace runs on. The two meet only at the contract. The workspace is therefore one instance of a reusable pattern: a domain-specific capture-and-trust system whose confirmed output is contract-expressible for consumption above it. Nothing in its design may assume it is the only workspace, or the top of the stack.

02 Design principles

These principles are extracted from the requirements and govern every behaviour that follows. They are the reason the workspace behaves the way it does, and they should outlast any individual feature.

CAPTURE AND DECISION-MAKING

→ **Operator records, PM decides.** The PM front-loads every decision into setup; the operator touches only what physically varies and what they observe. Field decision-making is slow and error-prone, so the architecture removes it from the field.

→ **AI proposes, a human approves, then it commits.** Plan import and test conversion propose; a person approves before anything is trusted. AI buys speed but never owns the record.

→ **Exceptions are declared, not inferred.** Redrill, test-fail, incident and near-miss are explicit human actions, never guessed from data. The system never invents a problem; accountability stays clear.

→ **Calculate the variances; declare the judgement calls.** The system computes grout and depth variance against design; humans declare safety and work exceptions. The machine does arithmetic; the human makes the call.

STATE, OWNERSHIP AND COMPLIANCE

→ **Status advances from the field, with PM override.** Operator actions move the work phase live; the PM can correct or set off-path states. The Status Board reflects reality without manual upkeep.

→ **A single owner at every state.** Exactly one role can edit a record at any review state, so a client deliverable never changes silently. Where a record drives payment or reaches a client, edits carry a light who-changed-what-when trail.

→ **Records are independent, not gated chains.** The natural daily sequence is encouraged, not enforced; gating is reserved for acts an individual can always complete alone.

→ **Gate only on acts an individual can always complete alone.** JSA sign-on gates drilling; nothing that needs another person ever gates. A compliance barrier must always be satisfiable by one person, and acknowledgement is individual and attributable, never applied in bulk.

VISIBILITY AND RELEASE

→ **Detect everything; release selectively.** The system surfaces every finding internally so nothing is missed; the PM decides which reach the client. Discretion for clarity, not concealment.

→ **Curate client exposure by the source of the issue.** Ground reality (lithology, conditions) tends toward the client; contractor performance (error, workmanship) is reviewed and resolved first.

→ **Inform, then act, then release.** Awareness, remediation and export are separated layers; the same exception is surfaced, resolved and released in sequence, never on one screen.

→ **Confirmed data only reaches the engineer.** The engineer and client never see in-progress or unreviewed work. This is the trust spine of the whole system.

ARCHITECTURAL INTENT

→ **A work type is a completion-criteria configuration on one spine.** The operational spine (Configure, Capture, Confirm) is invariant; drill, grout and test are the Capture content of the anchor (discrete-element) work type, and the type only defines which stages are required to submit. Only the confirmed record is released outward. The shared catalogue carries six ground-engineering work types (anchoring, piling and retaining, drilling, shotcrete, rockfall protection, drainage); V2.0 proves the spine in depth on anchoring and piling, the discrete-element pattern, and the other four onboard as configuration on the same spine. A new work type is a moved finish line, not new machinery.

→ **Set the conditions, don't build the feature.** Where a roadmap capability depends on data-model shape, the architecture keeps it reachable, a configuration or a zoom away, without building it now.

03 User model

Four roles operate the chain. A person may hold more than one (a supervisor is often also an operator), but the responsibilities, ownership and decision rights below are distinct. In ground engineering the engineer is the client, the commissioning party that relies on the record, so they form one external read tier, served by the PM's curated release.

ROLE	RESPONSIBILITIES	VISIBILITY	OWNERSHIP	DECISION RIGHTS
Operator	Capture field work: sign-on, drill / grout / test logs, lithology, and declared exceptions.	Own assigned anchors and in-progress logs, and the current JSA / SWMS.	Owns each log while it is In progress, until it is submitted for QA.	Declares what is observed and what went wrong (redrill, test-fail, incident); makes no setup or release decisions.
Supervisor	Submit the daily activity sheet; maintain the compliance and roster picture.	The project compliance picture and roster; own activity records.	Owns activity records (activity sheet edits carry a visible trail).	Decides what compliance and resourcing facts are recorded; cannot confirm or release work.
Project Manager	Set up projects, plan anchors, run QA, control client exposure, and release.	Everything: full project state, all logs, all exceptions, the QA queue.	Owns any record during QA, and owns the release decision; the only role that can re-open.	The single decision authority: confirms records, sets release, curates client exposure, overrides state.

ROLE	RESPONSIBILITIES	VISIBILITY	OWNERSHIP	DECISION RIGHTS
Engineer (client)	Rely on the confirmed record to make engineering judgements and track progress; trust it on sight.	Confirmed anchors only: the Status Board and the full confirmed detail, with curated exceptions.	Owns nothing in the workspace; consumes the trusted record.	None inside the system; engineering judgement is exercised outside it on a record it can trust on sight.

04 Workspace structure

The workspace presents a set of operational surfaces: areas where a defined consumer does a defined job. Described here by responsibility, not by layout: what each surface is for, who works in it, what it consumes and what it produces.

SURFACE	CONSUMER	OPERATIONAL RESPONSIBILITY	CONSUMES → PRODUCES
Landing / projects	All roles	The entry point: select a live project or create one; role determines what opens beyond it.	Identity, role → the project context for every other surface.
Setup & standards	PM	Front-load every decision: project, zones, specifications, work item templates, safety documents, plant library.	Client and scope info → a preset project the field inherits.
Work plan	PM	Place templates as planned anchors by zone and row; accept an AI-proposed plan from a drawing.	Templates, zones, plan drawing → planned anchors carrying their specs.
Project documents	PM · Operator	Hold inbound reference material (drawings, site instructions, lift plans, engineer-issued docs) as an offline project folder, distinct from JSA / SWMS and not a document-management system.	Uploaded reference files → an offline-available project reference the field reads (read-only).

SURFACE	CONSUMER	OPERATIONAL RESPONSIBILITY	CONSUMES → PRODUCES
Spatial map	Operator · PM	Locate work items on an uploaded site image (engineer drawing, photo or drone); plot by ID; give the operator a status-coloured map to identify exactly which item they are at.	Base image, planned items → a located, status-coloured map; tap-to-select into the drill log.
Sign-on & safety	Operator · Supervisor	Make compliance a fast, attributable, individual act; show who is outstanding on JSA sign-on.	Safety docs, sessions → daily sign-on and a live compliance picture.
Drill logs (capture)	Operator	Capture what physically happened, drill, grout, test, against the preset anchor, then submit.	Planned anchor, observations → a compiled anchor record advanced through its phases.
Incident management	Operator	Capture a safety event anywhere, anytime, in under a minute, with the critical facts protected.	Operator report → a standalone incident record and an immediate PM alert.
Daily activity	Supervisor	Record resourcing, crew and plant hours, for payroll and invoicing.	Signed-on roster, times → an auto-totalled activity sheet with an edit trail.
Overview (inform)	PM	See the project's state and everything needing attention within seconds; resolve nothing here.	All records, variance flags → awareness of progress, alerts and compliance.
Drill summary (act)	PM	Open submitted and flagged logs, apply the QA lens, remediate, and confirm.	Submitted logs, flags → confirmed anchors or hand-backs.
Reporting (release)	PM	Release a trusted deliverable and a curated client view; decide what reaches the client.	Confirmed anchors, exception decisions → export and the client progress view.

SURFACE	CONSUMER	OPERATIONAL RESPONSIBILITY	CONSUMES → PRODUCES
Engineer (client) view	Engineer (client)	Read the confirmed record: the Status Board (glance-legible) and the progressive detail.	Confirmed anchors → a trusted, near-real-time picture (read-only).
Settings	PM / admin	Hold the configuration the workspace runs on, including the work-type completion rules.	Configuration → the behaviour every other surface inherits.

Two status surfaces, one truth. The Status Board renders work items schematically by zone for tally and QA; the Spatial Map renders the same items located on a real site image so an operator on the ground can identify which item they are standing at. Both read the same work-phase state, and each work item carries an image-relative position so its pin persists with the plan.

05 Functional domains

Beneath the surfaces, the workspace is organised into eleven functional domains: the business-logic areas that own a capability. Each consumes defined inputs, produces defined outputs, and feeds the domains downstream of it.

DOMAIN	PURPOSE	INPUTS	OUTPUTS → DOWNSTREAM
Project setup	Front-load every decision so the field is preset.	Project, zones, specs, templates, safety docs, plan.	Planned anchors carrying template and specs → all capture domains.
Project documents	Hold inbound reference material as an offline project folder, not a DMS.	Uploaded drawings, site instructions, engineer-issued docs.	Offline-available reference set → field reference (read-only).
Spatial mapping	Locate work items on a site image for on-site identification.	Base image; planned items; image positions.	Located, status-coloured map → operator identification, drill-log selection.
Sign-on & safety	Make compliance a fast, attributable, individual act.	Safety docs; operator and supervisor sessions.	Sign-on and compliance picture → drilling (gate), reporting.

DOMAIN	PURPOSE	INPUTS	OUTPUTS → DOWNSTREAM
Drill logging	Capture what physically happened in the hole.	Planned anchor; lithology presets; observations.	Drilling record; phase → Drilled → grouting, QA.
Grouting	Capture the grout placed and any loss.	Drilled anchor; grout presets; bag counts.	Grouting record, derived volume, redrill flag → testing, QA, exceptions.
Testing	Digitise and distribute the paper test output.	Photo of the paper test record.	Verified test result; test-fail flag → QA, reporting.
QA & review	Turn captured work into a trusted record.	Submitted logs; system variance flags.	Confirmed anchors → engineer visibility, reporting; or hand-backs → capture.
Incident management	Capture safety events anywhere, fast.	Operator report: type, injury, severity, media.	Incident record + triage alert → PM review (internal only).
Daily activity	Record resourcing for payroll and invoicing.	Signed-on roster; times; plant library.	Activity sheet with auto-totals → export (cost stream).
Reporting	Release trusted progress outward.	Confirmed anchors; PM exception decisions.	Engineer (client) view + export → the engineer (client).

06 Core workflow architecture

The workspace runs one end-to-end pipeline, and every work type runs it as the same three-stage spine: Configure, Capture, Confirm, with the confirmed record released outward as the outcome of Confirm. Setup happens once; daily capture repeats per work item; QA and release run continuously alongside. Information only ever moves forward along this line, and ownership changes hands at two deliberate gates.

> THE INVARIANT SPINE · THE SAME FOR EVERY WORK TYPE
>
> **Configure → Capture → Confirm.** Release is not a fourth stage: only the confirmed record is released outward, as the outcome of Confirm.

THE ANCHOR INSTANCE · DRILL / GROUT / TEST IS THE CAPTURE CONTENT

Setup → Plan → Sign-on → Drill → Grout → Test → QA → Confirm → Report

TRANSITION	INFORMATION CREATED / CONSUMED	OWNERSHIP & WHY IT EXISTS
Setup → Plan	Consumes specs and templates; creates planned anchors that inherit them.	PM owns throughout. Specs must exist before anchors can inherit them, so the operator selects nothing in the field.
Plan → Sign-on	Creates the day's attributable compliance record.	Operator acts. Planned anchors are visible immediately, but drilling cannot begin until that day's JSA sign-on exists, the one enforced field gate.
Sign-on → Drill / Grout / Test	Creates the anchor's captured record; advances work phase.	Operator owns while In progress. The natural sequence, but each phase is an independent record that can be saved progressively.
Capture → QA	Consumes the compiled log; hands it to the PM.	Ownership passes to the PM; operator editing locks. Submission is allowed at any phase, enabling early escalation rather than only at close-out.
QA → Confirm	Creates the trusted record.	PM owns. Confirmation is the gate to the outside; only here does an anchor become engineer-visible. Re-opening reverses it and withdraws it from view.
Confirm → Report	Consumes confirmed anchors; creates the released deliverable.	PM owns release. Confirmed data is exposed progressively; reporting is the controlled release of a summary and export, curated by source.

Two gates, and only two. Sign-on gates drilling; confirmation gates external visibility. Everything else is an encouraged sequence, not an enforced chain, which is what lets a solo operator work without being blocked, and what lets the PM escalate or correct at any point.

07 State architecture

A work item carries two lifecycle dimensions that move through states, work-phase and review, with the testing flag and freshness riding alongside as independent overlays.

Conflating any of them into one field would stop the Status Board and the QA queue from telling the truth at the same time: an anchor can be Grouted (phase), still In progress (review) and flagged for testing, all at once.

WORK-PHASE LIFECYCLE · WHERE THE PHYSICAL WORK IS

Planned → Active → Complete.

The platform states are generic; the concrete phase is a read of which configured stages are complete, so the anchor reads as Planned, Drilled, Grouted, Tested. Work phase advances automatically as the operator closes each stage; the PM can override. Two off-path states sit outside the line: Requires-redrill (set by the operator's redrill flag, which also alerts the PM) and Not-required (a stage a work type does not need). Because a work type defines which stages are required, a valid pile reaches Complete at drilled.

REVIEW LIFECYCLE · WHO OWNS THE RECORD, AND WHETHER IT IS TRUSTED

In progress → Submitted for QA → Confirmed.

REVIEW STATE	WHO OWNS / CAN EDIT	EXTERNAL VISIBILITY
In progress	Operator (save at any stage).	None.
Submitted for QA	PM only (operator locked out).	None.
Confirmed	No one.	Engineer-visible.
Re-opened	PM only (reverts to QA).	Withdrawn until re-confirmed.

Re-open is the universal hand-back. It returns an anchor to review whether it was confirmed (fixing a mistake) or submitted early (now reviewed), releasing the edit lock and dropping it from client view until re-confirmed. A light who-and-when record is kept on every re-open and re-confirm.

TESTING AND FRESHNESS · INDEPENDENT OVERLAYS

Whether an anchor is load-tested is a separate boolean that moves on its own: not every anchor is tested, so the testing flag is neither a work phase nor a review state but an overlay. Freshness, how current the displayed status is, is the second overlay, computed from sync state. Keeping both independent is what lets the same spine serve work types that test and work types that do not, and lets an unrefreshed status read as ageing rather than as settled fact. The Technical & Data Architecture (§05) is the authority on this model.

DATA-MODEL NOTES

Identity. Each work item carries a stable system identifier, used by the ODC and any external reference, and a separate human reference such as B12, shown to users and re-labellable. Identity is not the label.

Derived project status. Above the per-item dimensions, a project carries a status derived from its work items (phase tallies, percent complete, open findings, QA backlog, last activity, a coarse state), computed and stored as a queryable property: the source of the Overview pulse and the ODC Status object.

Evidence. Captured photos, video and files are evidence items with provenance (who, when, source), attached to a parent record (work item, stage, incident, daily sheet). Keeping evidence a first-class attachable concept lets it generalise across work types without rework.

Project lifecycle. Separate from the derived status, a project carries an authored lifecycle flag (Open / Closed), set by the PM at closeout (backlog PLAT-11). Closed is declared, never computed, and travels on the ODC; the derived coarse state above remains system-computed.

08 Findings architecture

Findings are how the system handles work going other than planned. A finding is either a variance, computed by the system against design, or an exception, declared by a person; either is alerted to the PM, resolved, and released to the engineer at the PM's discretion by the source of the issue.

FINDING	DETECTION	ALERT	RESOLUTION	RELEASE
Requires redrill	Operator-declared (grout loss / failure).	Live PM alert on the anchor.	Manage redrill; set work phase.	Ground-driven; usually shared.
Test fail	Operator-declared at submission.	Live PM alert on the anchor.	Review test; remediate.	Reviewed by source before release.
Incident / near-miss	Operator-declared, standalone.	Immediate project-level alert.	Triage and review the event.	Internal safety record; not client-facing.
Grout variance	System-computed vs design.	Variance flag on Overview.	Review; annotate or remediate.	PM chooses whether to annotate.
Depth variance	System-computed vs design depth.	Variance flag on Overview.	Review; annotate or remediate.	PM chooses whether to annotate.

THE GOVERNING RULE

The system surfaces every finding, variance or exception, so nothing is missed, but the PM decides which reach the client. Ground reality tends toward the client because transparency builds trust; contractor performance is reviewed and resolved first. This is discretion for clarity and fairness, never concealment. Measurable variances are computed; judgement calls are declared by a person. An alert is the surfacing of either to the PM, not a third kind of finding.

09 Information visibility architecture

Visibility is owned, timed and reasoned. Each role sees a defined slice, at a defined moment, for a defined reason. Information progresses from private capture to trusted release in one direction only.

ROLE	WHAT THEY SEE	WHEN	WHY
Operator	Own assigned anchors and in-progress logs.	While editing, before submission.	Capture without exposure to others' work or judgement.
Supervisor	Compliance picture and roster across the project.	Live, through the day.	Turn individual records into a real-time compliance view.
PM	Everything: all logs, exceptions, variances, the QA queue.	Continuously.	The single point that sees all and decides what is released.
Engineer (client)	Confirmed anchors only: the Status Board and full technical detail, with curated exceptions.	Progressively, as each anchor is QA'd and the PM releases.	Trust the record on sight; never see unreviewed data or internal noise.

THE CONFIRMED-ONLY RULE

Nothing reaches the engineer (client) until a PM has confirmed it. In-progress and submitted work is invisible outside the team; re-opening a confirmed anchor immediately withdraws it from external view until re-confirmed. This single rule is what lets the engineer (client) trust the record on sight, and it is the boundary the whole visibility model exists to protect.

10 Reporting architecture

The PM's side of the workspace is three separated layers. The same finding is surfaced in one, resolved in the next, and released in the third, never collapsed onto a single screen. Information only ever moves down this stack; it never jumps straight from detection to the client.

LAYER	ROLE	WHAT MOVES THROUGH IT
Overview · inform	Surface the project's state and everything needing attention; resolve nothing.	Progress by phase and zone, alerts, variance flags, compliance: awareness within seconds.
Drill summary · act	Apply the QA lens: open flagged and submitted logs, remediate, confirm.	Anchor-by-anchor detail and the QA queue; variances and declared exceptions as filters in the same table.
Reporting · release	Release a trusted deliverable and push a curated client view.	Export of confirmed data; the PM controls whether exceptions are annotated or remediated first.

From detection to client. An issue is detected on Overview, resolved in Drill Summary, and only then released through Reporting. The separation is the architecture: it guarantees that nothing reaches the client that has not first been seen and acted on internally.

11 Emitting to the Operational Data Contract

The confirmed record is this workspace's output to the rest of the portfolio. The workspace emits it, and only it once confirmed, as a source-neutral summary conforming to the Operational Data Contract defined in the Product Portfolio Architecture §05. The workspace emits; it never reads. The mapping below is how the confirmed workspace record becomes the five contract objects. These five are the canonical object set; the Technical & Data Architecture maps the confirmed record into them rather than defining its own.

ODC OBJECT	WHAT THE WORKSPACE EMITS INTO IT · FROM THE CONFIRMED RECORD
Project Summary	The anchors job as one pin: name, organisation, location, overall status, percent complete (confirmed anchors / planned), open incident and active risk counts, PM and contractor.

ODC OBJECT	WHAT THE WORKSPACE EMITS INTO IT · FROM THE CONFIRMED RECORD
Status	The derived project status rolled up: Not started · Active · Blocked · Complete, with the phase breakdown (drilled / grouted / tested).
Progress	Anchors as the metric: planned total, completed, in progress, percent complete, straight from the Status Board.
Incident	Operator-declared incidents, near-misses and test-fails, surfaced upward once confirmed.
Risk	Computed grout and depth variance against design, plus declared ground conditions: the forward-looking concerns.

THE EMIT BOUNDARY

Only confirmed records are emitted: the confirmed-only-outward boundary travels onto the contract; nothing in-progress or unreviewed leaves the workspace. The workspace conforms to the contract; it does not define or version it. The contract is owned in the Portfolio Architecture §05 and held there as an assumed initial surface until a second workspace and a real consumer confirm it, so this emit shape is v0.1, to be exercised, not locked.

12 Future-proofing conditions

Five roadmap capabilities must remain reachable without being built now. For each, only the functional boundary that must stay intact is stated, never the design. These are day-one knowledge for the Technical Owner, because the one thing that cannot be cheaply changed later is the data model.

CAPABILITY	FUNCTIONAL BOUNDARY TO PRESERVE
Additional work types	A work type must remain a true completion-criteria configuration on the shared spine. The six-work-type ground-engineering catalogue (anchoring, piling and retaining, drilling, shotcrete, rockfall protection, drainage) onboards as configuration on the same spine; a type that is not a drill log at all must be addable without re-architecting, and the spine must not assume drill / grout / test.
Additional industry workspaces	The skeleton, capture → QA → confirmed record → curated client visibility, must stay domain-agnostic. Ground-engineering vocabulary (anchor, lithology, grout) lives in a domain layer above the shared workspace platform, never inside it.

CAPABILITY	FUNCTIONAL BOUNDARY TO PRESERVE
Operational Data Contract	The confirmed record maps to the ODC objects (see §11); keep it contract-expressible and do not build the contract itself: the contract is defined in the Portfolio Architecture §05. The workspace produces ground truth; the contract is the seam that carries it upward.
Coordination (read-side)	Every project must carry a geospatial location; an organisation tier must sit above projects; project status must be a first-class, queryable, aggregatable property exposed from outside the project; and the project must be addressable and enterable from outside itself.
Multi-contractor interoperability	The same source-neutral contract that feeds Coordination must be able to carry one contractor's confirmed record into another party's picture, without exposing internal or unconfirmed state. Trust boundaries travel with the data.

THE GOVERNING CONDITION

Build none of these in the workspace. Architect so that a new work type is a configuration, a new domain reuses the shared workspace platform, and Coordination reads an addressable, status-emitting project through the Operational Data Contract. Preserve the shape; defer the features. These boundaries are what keep this architecture valid even if the backlog changes substantially over the next two years.

13 Revision history

VERSION	NOTES
v1.6	Alignment to Requirements Backlog v0.17 and the decisions ledger. Project Reference Documents added as a surface (§04) and a functional domain (§05), bringing the domain count to eleven (backlog DOC-1 to 3). Work-phase lifecycle generalised to Planned, Active, Complete with the anchor read (Drilled, Grouted, Tested) shown as derived, matching the Technical & Data Architecture and backlog v0.17. Project lifecycle note added: an authored Open / Closed flag (backlog PLAT-11), distinct from the derived status. ODC Status emit aligned to the derived coarse tokens (Not started, Active, Blocked, Complete). Derivation updated to v0.17.
v1.5	State-model wording aligned to the Technical & Data Architecture (§05): two lifecycle dimensions (work-phase, review) with testing and freshness as independent overlays, replacing the earlier three-dimensions framing in §07. Wording only; no behavioural change.

VERSION	NOTES
v1.4	Alignment pass to Requirements Backlog v0.13–v0.15. Operational spine reframed to Configure / Capture / Confirm, with drill / grout / test as the Capture content of the anchor work type (resolving the §06–§12 contradiction). Anchor Board renamed Status Board and the state model generalised to the work item. Crew briefings and toolbox talks moved to roadmap; sign-on visibility reframed. Spatial Map added as a V2.0 surface and domain, with the Status Board / Spatial Map division of labour stated. §08 retitled Findings architecture with the variance / exception / alert taxonomy. Data-model notes added for work-item identity, derived project status and evidence (backlog PLAT-7 / 9 / 10). Derivation updated to v0.15.
v1.3	Reframe-alignment pass to the settled portfolio stack. §01 position rewritten to the two-tier model, a workspace on the TACEDGE Shared Workspace Platform that emits to the ODC; Coordination (Product A) is a separate product tier that reads the contract, not a layer above this workspace. Lowercase 'core' (the shared substrate) replaced with 'the shared workspace platform'. Derivation updated to Requirements Backlog v0.12. New §11 added: an explicit emit-mapping of the confirmed record onto the five ODC objects (emit-side conformance, mirroring the Coordination FA read-side). Future-proofing renumbered to §12.
v1.2	Merged Engineer and Client into one external read tier (engineer = client); Anchor Board and Site Map naming; coordination framing expressed through the ODC.
v1.0	First TACEDGE Ground Engineering Functional Architecture, the operating model between approved requirements and implementation.

PHASE 3 UX · THE COMPLETE SET · TWENTY-TWO V2.0 CORE SCREENS ACROSS THREE SURFACE GROUPS

Page Definition Sheets

The V2.0 core screen set for TACEDGE Ground Engineering: twenty-two sheets that prove the operational spine in depth on anchoring and piling, each defining the job a screen does for the person in front of it. It inherits the architecture rather than recreating it; the Functional Architecture, Requirements Backlog and Product Brief remain the source of truth.

STATUS	v1.2 · Complete (22 sheets)	CLASS	Internal
OWNER	Product Owner (founders)	PRODUCT	TACEDGE Ground Engineering (Workspace B1)
INHERITS	Ground Engineering FA v1.6 · Backlog v0.17 · Brief v3.2 (source of truth)	PHASE	3 · UX Design

The complete set

This is the Page Definition Sheet set for the TACEDGE Ground Engineering workspace: twenty-two screens, the V2.0 core set that proves the operational spine in depth on anchoring and piling. Each sheet defines what job it does for the person in front of it: its user question, information hierarchy, journey, inputs, outputs and success criteria.

It is written for UX and UI design, HTML-prototype generation and developer handover, and it inherits the architecture rather than recreating it: the Functional Architecture, Requirements Backlog and Product Brief remain the source of truth. Each sheet carries a one-line traceability footer back to the backlog.

Additional per-type capture screens for the other four catalogue work types, drilling, shotcrete, rockfall protection and drainage, extend these same patterns as configuration on the same spine, not as new sheets.

01 The sheet template

Every sheet carries the same eleven fields, ordered the way a designer reasons about a screen, question first, then hierarchy, journey and the in / out of information.

FIELD	WHAT IT CAPTURES
Header	Page · sheet ID · surface · primary user.
User Question	The single question this page exists to answer, the organising principle for the screen.
Purpose	One line of user intent: what the page is for.
Entry Points	How a user arrives, and any gate that must be passed first.
Information Hierarchy	Primary · Secondary · Tertiary, drives layout and visual weight.
User Journey	The normal click-path through the page, numbered.
Inputs	What information the page consumes.
Outputs	What information or decisions the page produces.
Key Interactions	The few actions that define the screen for design.
Success Criteria	How we know the screen works, aligned to the Brief's Definition of Done.
Screen-specific rules	Only the architectural behaviour that visibly changes this page; the rest is inherited.

02 The page inventory

Twenty-two sheets across three surface groups, each grounded in a requirement family. The PM surfaces run on the Configure, Capture, Confirm spine.

PAGE	SHEET ID	PRIMARY USER
A · Field & Operator Surfaces		
Drill Log · Anchor	`PDS-DRL`	Operator

PAGE	SHEET ID	PRIMARY USER
Crew Sign-On	PDS-SON	Operator · Supervisor
Piling Log	PDS-PIL	Operator
Grout Log	PDS-GRT	Operator
Anchor Test Record	PDS-TST	Operator · Supervisor
Incident / Near-Miss	PDS-INC	Any user
Daily Activity Sheet	PDS-DAS	Supervisor · Operator
Safety Docs (JSA / SWMS)	PDS-SAF	Operator · Supervisor
B · PM Setup Surfaces		
Project Home	PDS-OVW	Project Manager
Layout	PDS-LAY	Project Manager
Work Item Template	PDS-DSG	Project Manager
Work Plan	PDS-PLN	Project Manager
Testing Standards	PDS-STD	Project Manager
Evidence & QA Requirements	PDS-QAR	Project Manager
Project Reference Documents	PDS-DOC	Project Manager · Operator
Safety & Risk Controls	PDS-SRC	PM · Supervisor
C · Monitoring & External Surfaces		
Status Board · Overview	PDS-BRD	Project Manager
Spatial Map	PDS-SPM	Project Manager · Operator
Drill Summary · QA Queue	PDS-QAQ	Project Manager
Reporting	PDS-RPT	Project Manager

PAGE	SHEET ID	PRIMARY USER
Project Closeout	PDS-CLO	Project Manager
Engineer (Client) View	PDS-ENG	Engineer (client)

SECTION A · FIELD & OPERATOR SURFACES

Eight screens · the field captures, the operator's world.

PDS-DRL Drill Log · Anchor

Surface. Field · Operator capture. **Primary user.** Operator (driller).

User question. "Which anchor am I on, and what do I record about it?"

Purpose. The operator's point-of-drilling capture of a single anchor, the screen that replaces the paper logbook. The make-or-break adoption surface: a cold, gloved operator must begin capture in seconds.

ENTRY POINTS

→ From the operator's anchor list for the day, or by descending from a Status Board pin, or by tapping a pin on the Spatial Map (SPM-4).

→ Gated: drilling is unavailable until the operator has personally completed JSA sign-on for the day. If outstanding, the page leads with the sign-on prompt, not the capture fields.

INFORMATION HIERARCHY

Primary. This anchor's identity (zone · ID) and the one field being captured now, actual depth, then lithology. The act of recording dominates the screen. **Secondary.** Inherited design context, design depth, specification, lithology expectation, visible for comparison without leaving the page. **Tertiary.** Prior entries for this anchor and full history, one tap away, off the capture path.

USER JOURNEY

→ Opens the anchor from the day's list (or arrives gated to sign-on first).

→ Glances at inherited design context to orient.

→ Records actual depth, then lithology, then any observation, typed, or by voice.

→ If something went wrong, declares an exception (redrill / test-fail) in one action.

→ Submits the anchor for QA. The work phase advances on its own.

Inputs. Anchor design (zone, ID, design depth, specification, lithology expectation), set by the PM in setup · the operator's JSA sign-on state for the day.

Outputs. A captured drill record for the anchor · an optional operator-declared exception that alerts the PM · a submitted-for-QA anchor.

KEY INTERACTIONS

- → Record depth / lithology / observation, typed baseline, voice as a Should.
- → Declare exception. A single, prominent action; never buried, never inferred.
- → Submit for QA. Done, or needs-your-eyes.

SUCCESS CRITERIA

- → A cold operator begins capture within seconds, no training.
- → Faster than the paper logbook it replaces.
- → Declaring an exception is as fast as recording a normal anchor.

SCREEN-SPECIFIC RULES

- → The operator records what varies and observes; everything else is inherited from setup and shown read-only, so the screen stays sparse.
- → Offline & back-fill must be legible: a back-dated or offline-captured entry is visibly marked as such, so it never reads as live (DRL-5).

Traceability. `DRL-1 · DRL-2 · DRL-3 · DRL-4 · DRL-5 · DRL-6 · QA-1 · QA-3 · SAF-1 · ALT-1`

PDS-SON Crew Sign-On

Surface. Field · Compliance capture. **Primary user.** Operator · Supervisor.

User question. "Am I signed on and cleared to work today?"

Purpose. The attributable, individual sign-on that clears a person to work on this project today, and the gate that unlocks drilling. Replaces the paper sign-on sheet.

ENTRY POINTS

- → Start of day, from the project, the first thing prompted on arrival.
- → It is the gate: until a person signs on, gated work (drilling) is unavailable to them.

INFORMATION HIERARCHY

Primary. The sign-on action itself, who I am, this project, today, and my cleared / not-cleared state. **Secondary.** The JSA / SWMS I am acknowledging by signing on, available to read here. **Tertiary.** Today's crew already signed on, roster context, one glance away.

USER JOURNEY

- → Arrives at the project for the day.
- → Reviews the JSA / SWMS governing today's work.

→ Signs on through their own authenticated session.

→ Sign-on clears them; gated work unlocks.

Inputs. The project's active JSA / SWMS · the person's own identity and session.

Outputs. An attributable sign-on record · the unlock of gated work (drilling) for that person, that day.

KEY INTERACTIONS

→ Sign on. A single, individual, attributable action.

→ Acknowledge the JSA / SWMS as part of signing on.

SUCCESS CRITERIA

→ Crew sign-on completes in under 10 seconds (Brief Tier-1).

→ A solo operator can sign themselves on, alone.

→ Cannot be applied in bulk by a supervisor on others' behalf.

SCREEN-SPECIFIC RULES

→ Individual and attributable: sign-on is made through the person's own session, there is no 'sign all' control on this screen.

→ Records are independent, sign-on gates drilling and waits on nothing else.

Traceability. `SOS-1` · `SOS-2` · `SOS-3` · `SOS-4` · `SAF-1`

PDS-PIL Piling Log

Surface. Field · Operator capture. **Primary user.** Operator (driller).

User question. "Which pile am I on, and what do I record about it?"

Purpose. Point-of-work capture of a single pile, the piling sibling of the Drill Log, on the same shared spine with a different required-phase set.

ENTRY POINTS

→ From the operator's pile list for the day, or by descending from a Status Board pin, or by tapping a pin on the Spatial Map (SPM-4).

→ Gated by JSA sign-on for the day, exactly as the Drill Log.

INFORMATION HIERARCHY

Primary. This pile's identity (zone · ID) and the field being captured now. **Secondary.** Inherited design context, design parameters, specification, for comparison. **Tertiary.** Prior entries for this pile and full history, off the capture path.

USER JOURNEY

→ Opens the pile from the day's list.

- → Glances at inherited design context to orient.
- → Records the captured values, typed or by voice.
- → Declares an exception if something went wrong.
- → Submits the pile for QA; the work phase advances on its own.

Inputs. Pile design (front-loaded by the PM in setup) · the operator's sign-on state.

Outputs. A captured pile record · an optional operator-declared exception · a submitted-for-QA pile.

KEY INTERACTIONS

- → Record the captured values, typed baseline, voice as a Should.
- → Declare exception. A single, prominent action.
- → Submit for QA.

SUCCESS CRITERIA

- → A cold operator begins capture in seconds, no training, the same adoption bar as the Drill Log.
- → Faster than the paper logbook it replaces.

SCREEN-SPECIFIC RULES

- → Work type is a configuration on one spine: piling is the same screen as the drill log with a different done-line, not new machinery (Brief DoD 6, Platform). The same spine carries the full six-type catalogue, anchoring, piling and retaining, drilling, shotcrete, rockfall protection and drainage; V2.0 proves it in depth on anchoring and piling.

Traceability. `PIL-1` · `PIL-2` · `PIL-3` · `DRL-3` · `QA-1`

PDS-GRT Grout Log

Surface. Field · Operator capture. **Primary user.** Operator (driller).

User question. "How much grout went in, and does it match design?"

Purpose. Capture of the grouting phase for an anchor, the recorded grout, with variance against design computed for the operator, not calculated by them.

ENTRY POINTS

- → From the anchor's grout phase, on the operator's list or by descending from a pin.
- → Gated by JSA sign-on for the day.

INFORMATION HIERARCHY

Primary. The grout capture for this anchor, the values being recorded now. **Secondary.** The design expectation and the computed variance flag against it. **Tertiary.** Prior grout entries for the anchor, off the capture path.

USER JOURNEY

→ Opens the anchor's grout phase.

→ Records the grout placed.

→ The system computes variance against design and surfaces a flag.

→ Submits for QA.

Inputs. Anchor design (grout design parameters) · the anchor's drill record.

Outputs. A captured grout record · a system-computed grout / depth variance · a submitted-for-QA anchor.

KEY INTERACTIONS

→ Record grout placed.

→ Review the computed variance flag.

→ Submit for QA.

SUCCESS CRITERIA

→ Capture is fast; the operator never calculates variance by hand.

→ Out-of-tolerance grout is unmistakable at a glance.

SCREEN-SPECIFIC RULES

→ Calculate the measurable, declare the judgement: grout / depth variance is system-computed against design and shown here as a flag, not a field the operator fills, and not operator-declared (contrast redrill / test-fail).

Traceability. `GRT-1` · `GRT-2` · `GRT-3` · `GRT-7` · `GRT-9` · `QA-1`

PDS-TST Anchor Test Record

Surface. Field · Capture + AI convert. **Primary user.** Operator · Supervisor.

User question. "Did this anchor pass its test, and is the result captured cleanly?"

Purpose. Captures the anchor test result by digitising the paper test output through AI conversion, human-verified. The value is downstream digitising, not re-keying at the gauge.

ENTRY POINTS

→ From the anchor, once tested on the gauge.

→ Reached when a test is due against the project's testing standard.

INFORMATION HIERARCHY

Primary. The test outcome, pass / fail, and the captured or AI-converted figures. **Secondary.** The testing standard (what a pass is) and the source photo the figures came from. **Tertiary.** Prior tests for the anchor, off the path.

USER JOURNEY

→ Test is performed on paper at the gauge.

→ Photographs / uploads the test sheet.

→ AI proposes the extracted figures.

→ Operator or PM verifies the figures before commit.

→ Result is recorded; pass / fail is set, and a fail raises an exception.

Inputs. The paper test output (photo) · the testing standard (pass criteria) · the anchor's identity.

Outputs. A verified digital test record · a pass / fail outcome · an operator-declared test-fail exception if failed.

KEY INTERACTIONS

→ Capture / upload the test sheet.

→ Verify the AI-proposed figures before commit.

→ Confirm pass / fail.

SUCCESS CRITERIA

→ AI conversion is accurate enough that verification is glance-fast.

→ Paper still wins at the gauge; nothing is double-keyed.

SCREEN-SPECIFIC RULES

→ AI proposes, human approves, then commits: nothing is trusted until the operator or PM verifies the extracted data (TST-3).

→ A test-fail is operator-declared and alerts the PM.

Traceability. `TST-1` · `TST-2` · `TST-3` · `TST-4` · `ALT-2`

PDS-INC Incident / Near-Miss

Surface. Field · Standalone capture. **Primary user.** Any user.

User question. "Something happened, how do I report it fast?"

Purpose. Standalone capture of an incident or near-miss in under a minute, alerting the PM on submission. Independent of any other record, available to anyone, anywhere.

ENTRY POINTS

- → From anywhere in the workspace, a standing, always-available action.
- → Never gated; never waits on another record.

INFORMATION HIERARCHY

Primary. The fast capture, what happened, severity, photo or video. **Secondary.** Optional location / anchor context, if relevant. **Tertiary.** Prior incidents on the project, off the path.

USER JOURNEY

- → Taps incident / near-miss from anywhere.
- → Captures what happened, with photo or video.
- → Sets severity on the standard scale.
- → Submits; the PM is alerted immediately.

Inputs. The reporter's account · optional photo or video · the severity scale.

Outputs. An incident / near-miss record · an immediate PM alert.

KEY INTERACTIONS

- → Capture. Fast, photo-first.
- → Set severity on the locked scale.
- → Submit.

SUCCESS CRITERIA

- → Submission completes in 30–60 seconds (Brief Tier-1).
- → Capture succeeds instantly on-device even offline, syncing later.
- → Available to any user, from anywhere, with no setup.

SCREEN-SPECIFIC RULES

- → Standalone and operator-declared: never inferred, never gated on another record; one action alerts the PM.
- → Offline-first: photo / video capture must succeed on-device immediately, the operator thesis depends on it.

Traceability. `INC-1` · `INC-2` · `INC-3` · `ALT-1`

PDS-DAS Daily Activity Sheet

Surface. Field · Daily capture. **Primary user.** Supervisor · Operator.

User question. "Who and what plant was on site today, for how long?"

Purpose. The daily resourcing sheet, crew and plant hours, auto-totalled and exported. It captures the hours that drive pay and invoicing; the computation itself is roadmap.

ENTRY POINTS

→ From the project, for the current day.

→ Opened daily by the supervisor; no gate.

INFORMATION HIERARCHY

Primary. Today's crew and plant rows, with hours against each. **Secondary.** The auto-computed totals. **Tertiary.** Prior days' sheets, one click away.

USER JOURNEY

→ Opens today's sheet.

→ Adds crew and plant, each with hours.

→ Totals compute automatically.

→ Exports or submits the day.

Inputs. The crew roster · the plant list · hours worked.

Outputs. A daily resourcing record with auto-totalled hours · an export.

KEY INTERACTIONS

→ Add crew and plant rows.

→ Enter hours; totals compute.

→ Export the day.

SUCCESS CRITERIA

→ Hours auto-total, no manual sums.

→ The day exports cleanly with zero re-entry.

SCREEN-SPECIFIC RULES

→ This is money-facing data: edits carry a light who-changed-what-when trail (DAS-6).

→ Pay / invoice computation is out of scope, V2.0 captures and exports hours only. The project plant library is defined once in setup (DAS-5) and added on site by the supervisor.

Traceability. `DAS-1` · `DAS-2` · `DAS-3` · `DAS-4` · `DAS-5` · `DAS-6`

PDS-SAF Safety Docs (JSA / SWMS)

Surface. Field · Read & acknowledge. **Primary user.** Operator · Supervisor.

User question. "What safety documents govern today's work, and have I acknowledged them?"

Purpose. The field-side view of the project's JSA / SWMS, uploaded, visible, acknowledged. The acknowledgement feeds the sign-on gate; in-product authoring is roadmap.

ENTRY POINTS

→ From the project, or reached through sign-on (the JSA is acknowledged there).

→ Always available to read during the work.

INFORMATION HIERARCHY

Primary. The active JSA / SWMS for this work, and my acknowledgement state. **Secondary.** Upload / replace the document set (supervisor · PM). **Tertiary.** Prior versions of the documents, off the path.

USER JOURNEY

→ Opens safety docs, or reaches them via sign-on.

→ Reads the active JSA / SWMS.

→ Acknowledges, feeding the sign-on gate.

→ Supervisor / PM uploads or replaces documents as needed.

Inputs. The uploaded JSA / SWMS · the person's identity.

Outputs. An acknowledgement record that feeds the sign-on gate · the visible safety-document set.

KEY INTERACTIONS

→ Read the active document.

→ Acknowledge. Individual and attributable.

→ Upload / replace (supervisor · PM).

SUCCESS CRITERIA

→ The current JSA / SWMS is unmistakable and one tap to read.

→ Acknowledgement is individual and attributable, and it gates drilling.

SCREEN-SPECIFIC RULES

→ Acknowledgement feeds the JSA sign-on gate that unlocks drilling.

→ In-product authoring is out of scope, upload only in V2.0.

Traceability. `SAF-1` · `SOS-1`

SECTION B · PM SETUP SURFACES

Eight screens · standing a project up.

PDS-OVW Project Home

Surface. PM · Project hub. **Primary user.** Project Manager.

User question. "What does this project need from me right now?"

Purpose. The PM's home for a project, the front door that surfaces what needs attention and routes into every area. During setup it shows what is still to configure; during delivery it shows what needs action.

ENTRY POINTS

→ The PM's landing when opening a project, and the default tab.

→ Reachable from anywhere in the project via the breadcrumb.

INFORMATION HIERARCHY

Primary. What needs attention now, setup gaps, anchors awaiting QA, open alerts and incidents, anything blocking progress. **Secondary.** The project's headline state (status · progress) and the navigation into each area, Layout, Work Item Template, Work Plan, the Status Board, Reporting. **Tertiary.** Project metadata and settings, client, location, team, one click away.

USER JOURNEY

→ PM opens the project and lands here.

→ Scans what needs attention now.

→ Acts on the top item, or navigates into the relevant area.

→ Returns here as the project's home between tasks.

Inputs. Setup completeness · live status, progress, open alerts and QA-queue depth, consumed from across the project.

Outputs. Navigation and decisions, it routes the PM to the right screen. It produces no records of its own.

KEY INTERACTIONS

→ Act on an attention item.

→ Navigate into a project area.

→ Jump to an unfinished setup sheet (during setup).

SUCCESS CRITERIA

→ A PM understands what the project needs within 10 seconds of landing (Brief: PM status on login).

→ A cold PM can reach every setup area from here without hunting.

→ Nothing important is more than one tap away.

SCREEN-SPECIFIC RULES

→ A hub, not a capture screen: it informs and routes; it never records field truth.

→ It surfaces attention by source, setup gaps, delivery alerts and QA are distinguishable, not merged into one list.

Traceability. `SET-1 · SET-2 · SET-3 · PMV-1 · PMV-2`

PDS-LAY Layout

Surface. PM · Setup. **Primary user.** Project Manager.

User question. "How is the site divided into zones?"

Purpose. Where the PM defines the site's zones, the spatial divisions the whole project is organised by. Anchors, the board and the operator's day list all key off zones, so this is set early.

ENTRY POINTS

→ From the Project Home hub or the setup flow.

→ An early step, zones precede the Work Plan, which places anchors into them.

INFORMATION HIERARCHY

Primary. The zones, naming and dividing the site. The act of structuring space dominates the screen. **Secondary.** Zone properties (name, sequence, attributes) and a count of how many. **Tertiary.** The underlying site reference (a plan or image, if uploaded), context only.

USER JOURNEY

→ PM opens Layout, early in setup.

→ Creates zones, naming and ordering them.

→ Adjusts zone properties.

→ Saves; zones are now available to the Work Plan and the board.

Inputs. An optional site plan or reference · the PM's knowledge of how the site divides.

Outputs. The set of zones the project is organised by, consumed by the Work Plan, the Status Board and every operator capture page.

KEY INTERACTIONS

→ Create / name a zone.

→ Order the zones.

→ Edit zone properties.

SUCCESS CRITERIA

→ A PM defines the site's zones in minutes, without a rebuild later.

→ Zones map clearly to how the site is actually run.

→ The Work Plan can immediately place work into them.

SCREEN-SPECIFIC RULES

→ Foundational: zones are set early because the plan, the board and capture all key off them; changing them later ripples downstream.

→ Schematic + spatial in V2.0: located plotting on an uploaded image is the Spatial Map; full real-space layout on image or LiDAR is the Site Map roadmap item.

Traceability. SET-2

PDS-DSG Work Item Template

Surface. PM · Reference library. **Primary user.** Project Manager.

User question. "What anchor specifications exist for this project?"

Purpose. The project's library of anchor specifications, the design types (depth, load, grout and test spec) that planned anchors are assigned from. Authored once and reused across the plan.

ENTRY POINTS

→ From the Project Home hub or the setup flow.

→ Precedes the Work Plan, designs must exist before they can be assigned to planned anchors.

INFORMATION HIERARCHY

Primary. The list of anchor designs for this project, the reference library. **Secondary.** Each design's parameters, design depth, load, grout and test spec. **Tertiary.** Where each design is used (which planned anchors reference it), one click away.

USER JOURNEY

→ PM opens Work Item Template in setup.

→ Creates a design type with its parameters.

→ Repeats for each spec the project uses.

→ Saves; designs are now assignable in the Work Plan.

Inputs. The engineer's specification and design intent for the project.

Outputs. A reusable library of anchor designs, assigned to planned anchors in the Work Plan, and inherited as read-only context on the operator's capture pages.

KEY INTERACTIONS

→ Create a design.

→ Set its parameters.

→ Edit / duplicate a design.

SUCCESS CRITERIA

→ A PM authors the project's specs once and reuses them, never re-entering parameters per anchor.

→ A design's parameters flow automatically into every anchor that uses it.

SCREEN-SPECIFIC RULES

→ A reference library, not a capture screen: designs are authored once and referenced everywhere; the operator inherits them read-only and never edits them.

→ Distinct from the Work Plan, the Work Item Template defines the specs; the Plan places the work built to them.

Traceability. `DSN-1 · DSN-2 · DSN-3 · SPEC-1 · SPEC-2 · SPEC-3 · SPEC-4`

PDS-PLN Work Plan

Surface. PM · Spatial authoring. **Primary user.** Project Manager.

User question. "What work needs to occur, and where?"

Purpose. Where the PM lays out the planned work, placing the anchors to be drilled into the site's zones, producing the plan the field executes against. AI proposes from an imported PDF plan; the PM approves and adjusts.

ENTRY POINTS

→ From the Project Home hub or the setup flow, once Layout (zones) and Work Item Template (specs) exist.

→ A PDF plan import can be started here.

INFORMATION HIERARCHY

Primary. The plan canvas, zones with planned anchors placed in them. The act of placing and adjusting work dominates the screen. **Secondary.** The anchor designs available to assign, and counts per zone (planned vs target). **Tertiary.** The source PDF plan and import history, off the canvas.

USER JOURNEY

→ PM opens the Work Plan, zones already defined in Layout.

→ Imports a PDF plan; AI proposes zones and anchor counts.

→ Reviews and approves the proposal, or places anchors by hand.

- → Assigns an anchor design to the planned anchors.
- → Saves the plan; the field now has work to execute.

Inputs. The site Layout (zones) · the Work Item Templates (specs) · an optional imported PDF plan.

Outputs. A planned anchor layout, the anchors to be drilled, placed in zones, with designs assigned. The source of the operator's day list and the Status Board's pins.

KEY INTERACTIONS

- → Import plan. PDF to AI proposal.
- → Approve / adjust the proposal.
- → Place / move a planned anchor.
- → Assign a design.

SUCCESS CRITERIA

- → A PM creates a complete plan without manually creating every anchor, the import does the bulk.
- → The proposal is accurate enough that approving is faster than drawing.
- → The plan reads clearly by zone.

SCREEN-SPECIFIC RULES

- → AI proposes, human approves, then commits: the PDF import proposes zones and counts; nothing is planned until the PM approves (PLAN-2).
- → Schematic + spatial in V2.0: the plan is zone-based, plus located plotting on an uploaded image (the Spatial Map); full real-space placement on image or LiDAR is the Site Map roadmap item.

Traceability. `PLAN-1 · PLAN-2 · PLAN-3`

PDS-STD Testing Standards

Surface. PM · Setup / reference. **Primary user.** Project Manager.

User question. "What does a pass look like?"

Purpose. Where the PM sets the testing standards for the project, the criteria that define a passing anchor test. The Anchor Test Record references these so a pass or fail is declared against a known bar.

ENTRY POINTS

- → From the Project Home hub or the setup flow.
- → Set before testing begins.

INFORMATION HIERARCHY

Primary. The pass criteria, the thresholds a test must meet. **Secondary.** Which anchor designs each standard applies to. **Tertiary.** Reference to the governing engineering standard, context only.

USER JOURNEY

→ PM opens Testing Standards in setup.

→ Defines the pass criteria (thresholds).

→ Maps each standard to the relevant anchor designs.

→ Saves; the Test Record now shows these as the pass bar.

Inputs. The engineer's / governing testing requirements · the anchor designs.

Outputs. The pass criteria the Anchor Test Record declares pass or fail against, driving the pass / fail outcome and the test-fail exception.

KEY INTERACTIONS

→ Define a pass threshold.

→ Map a standard to designs.

→ Edit a standard.

SUCCESS CRITERIA

→ A PM sets the pass bar once; every test is declared against the same explicit bar, with no re-deciding in the field.

→ The standard is unambiguous enough that a fail is never a judgement argument.

SCREEN-SPECIFIC RULES

→ Defines what 'pass' means: the Anchor Test Record shows these criteria so the operator declares pass or fail against them; automatic evaluation against the standard is roadmap, not V2.0.

→ Pairs with the Anchor Test Record, which is the capture and conversion side.

Traceability. TST-2 · TST-3 · TST-4

PDS-QAR Evidence & QA Requirements

Surface. PM · Setup. **Primary user.** Project Manager.

User question. "What must be captured for a record to be trusted?"

Purpose. Where the PM sets what evidence a record must carry to be confirmable, the requirements that define a complete, trustworthy anchor record before it can be released to the client.

ENTRY POINTS

- → From the Project Home hub or the setup flow.
- → Set during setup; governs the QA queue downstream.

INFORMATION HIERARCHY

Primary. The evidence requirements, what each record must include to be confirmable (photos, test result, sign-off). **Secondary.** Which work types and designs each requirement applies to. **Tertiary.** The rationale or client expectation behind a requirement, context only.

USER JOURNEY

- → PM opens Evidence & QA Requirements in setup.
- → Defines what a complete record must carry.
- → Maps requirements to work types.
- → Saves; the QA queue now checks records against these before confirmation.

Inputs. The client's / engineer's evidence expectations · the work types and designs.

Outputs. The completeness rules the QA queue enforces, what makes a record confirmable and releasable.

KEY INTERACTIONS

- → Define an evidence requirement.
- → Map it to work types.
- → Edit a requirement.

SUCCESS CRITERIA

- → A PM sets the evidence bar once; the QA queue enforces it automatically, so nothing reaches the client under-evidenced.
- → A record's readiness for confirmation is unambiguous.

SCREEN-SPECIFIC RULES

- → Sets the confirmable bar: the QA queue and the confirmed-only-outward gate enforce these requirements, this screen sets the rule, the QA Queue applies it.
- → Inherits the FA's review lifecycle; it does not restate it.

Traceability. `QA-1` · `QA-2` · `QA-3`

PDS-DOC Project Reference Documents

Surface. PM · Setup / reference. **Primary user.** Project Manager · Operator.

User question. "What reference material does this job run on, and is it on my device?"

Purpose. A bounded project folder for inbound reference material (drawings, site instructions, lift plans, engineer-issued docs), available offline, distinct from the JSA / SWMS sign-on gate and not a document-management system.

ENTRY POINTS

→ From the Project Home hub or the setup flow, where the PM uploads and categorises the reference set.

→ Reached from the project on the operator device, read-only and available offline.

INFORMATION HIERARCHY

Primary. The reference documents themselves, categorised and openable, and whether each is available offline on this device. **Secondary.** Upload and categorise (the PM), with file type and size. **Tertiary.** How this differs from the JSA / SWMS gate, context only.

USER JOURNEY

→ PM uploads the reference documents in setup and categorises them.

→ Operator opens the project on their device.

→ Opens a drawing or instruction offline, without hunting.

→ No acknowledgement and no gate; reference only.

Inputs. Uploaded reference files (drawings, instructions, lift plans, engineer-issued docs) · their categories.

Outputs. An offline-available project reference set the field reads.

KEY INTERACTIONS

→ Upload and categorise a reference document (PM).

→ Open and view a document (all roles), offline.

→ No version control in V2.0; it is a folder, not a DMS.

SUCCESS CRITERIA

→ The field opens the drawings a job runs on, offline, without hunting or chasing.

→ The reference set is clearly distinct from the sign-on gate.

SCREEN-SPECIFIC RULES

→ A folder, not a DMS: no version control, approval workflow or in-app editing; it reuses the platform evidence and file model.

→ Acknowledgement and the drilling gate stay with sign-on (SOS); this sheet never gates work.

Traceability. DOC-1 · DOC-2 · DOC-3

PDS-SRC Safety & Risk Controls

Surface. PM · Setup. **Primary user.** Project Manager · Supervisor.

User question. "What safety documents govern this work?"

Purpose. Where the PM or supervisor sets up the project's safety documents, the JSA / SWMS that govern the work and that the field acknowledges. The authoring side of the field's Safety Docs page.

ENTRY POINTS

→ From the Project Home hub or the setup flow.

→ Set before work begins, the JSA gates drilling.

INFORMATION HIERARCHY

Primary. The project's governing safety documents, what is uploaded and active. **Secondary.** Which work or zones each document governs, and acknowledgement coverage. **Tertiary.** Prior versions and history, off the path.

USER JOURNEY

→ PM or supervisor opens Safety & Risk Controls in setup.

→ Uploads the JSA / SWMS governing the work.

→ Sets what each document governs.

→ Saves; the documents are live for field acknowledgement and gate drilling.

Inputs. The JSA / SWMS documents · the work or zones they govern.

Outputs. The active safety-document set the field views and acknowledges, feeding the sign-on gate.

KEY INTERACTIONS

→ Upload a document.

→ Set its scope.

→ Replace / version a document.

SUCCESS CRITERIA

→ The right safety docs are in front of the field before work starts.

→ Acknowledgement coverage is visible at a glance.

→ The gate, no sign-on without the JSA, is reliably wired.

SCREEN-SPECIFIC RULES

→ The authoring side: the field's Safety Docs page is the read-and-acknowledge counterpart; the active JSA feeds the sign-on gate.

→ In-product authoring of the documents themselves is roadmap, V2.0 uploads and manages.

Traceability. SAF-1 · SOS-1

SECTION C · MONITORING & EXTERNAL SURFACES

Six screens · Configure, Capture, Confirm, the located map and the engineer view.

PDS-BRD Status Board · Overview

Surface. PM · Live status board. **Primary user.** Project Manager.

User question. "Where does every anchor stand right now?"

Purpose. The live status board, every planned anchor placed by zone, showing its state at a glance. The inform layer of the PM's Configure, Capture, Confirm flow: where the project stands, read in seconds.

ENTRY POINTS

→ A primary project tab, and where the PM lands to read status.

→ Click-pin-descend: selecting an anchor descends into its log or detail.

INFORMATION HIERARCHY

Primary. Every anchor's state across the board, by zone, colour and state legible in seconds. The glance-first read dominates. **Secondary.** The headline roll-up, counts by state, progress (done / total), open alerts. **Tertiary.** Filters, and a single anchor's detail, one click or descent away.

USER JOURNEY

→ PM lands on the board, from the hub or on login.

→ Reads the whole project's state at a glance, planned, active, complete, and anything in issue.

→ Spots anything in issue or alerting.

→ Clicks a pin to descend into that anchor.

Inputs. The planned anchor layout (from the Work Plan) · the live state of each anchor (work-phase · review · testing · freshness) · open alerts and incidents.

Outputs. Situational awareness and navigation, the PM's read of where things stand, and the descent into any anchor. It produces no records.

KEY INTERACTIONS

→ Read the board at a glance.

→ Filter by zone or state.

→ Click-pin-descend into an anchor.

SUCCESS CRITERIA

→ A PM understands project status within 10 seconds of landing (Brief Tier-1).

→ The state of every anchor is legible without opening it.

→ Anything in issue is immediately obvious.

SCREEN-SPECIFIC RULES

→ The inform layer: it shows status, it does not resolve it, the Drill Summary · QA Queue is where the PM acts.

→ Glance-first: every element earns its place by being readable in seconds; the board reads the live lifecycle state directly and never re-computes truth.

→ Schematic + spatial in V2.0: by zone, with the located plot on an uploaded image as the Spatial Map; full geospatial or LiDAR placement is the Site Map roadmap item.

Traceability. `PMV-1 · PMV-2 · PMV-3 · PMV-4 · ALT-1`

PDS-SPM Spatial Map

Surface. Operator · PM · located map. **Primary user.** Project Manager · Operator.

User question. "Where on site is this anchor, and which one am I at?"

Purpose. The located status map, every planned anchor plotted on an uploaded site image (engineer drawing, photo or drone), coloured by work phase. The operator's on-site identification surface and the PM's located oversight: which physical anchor is which, read at a glance.

ENTRY POINTS

→ A primary project tab for the PM, and the operator's on-site map for the day.

→ Tap to select: the operator taps a pin to select that anchor and descend into its drill log (SPM-4); the PM into its detail.

INFORMATION HIERARCHY

Primary. The site image with every anchor plotted and coloured by work phase, which pin is which, located in real space. The on-site identification read dominates. **Secondary.** The selected anchor's identity (its B12-style reference) and state, and the operator's position relative to it. **Tertiary.** Zone filters, and the schematic Status Board one tap away, the same anchors as a grid.

USER JOURNEY

→ Operator opens the site map for the day, or the PM lands on it from the hub.

→ Sees the anchors plotted on the site image, coloured by phase.

→ Identifies the physical anchor in front of them by its pin, is this B12 or B13?

→ Taps the pin to descend: the operator into the drill log, the PM into the detail.

Inputs. The uploaded site base image and plotted anchor positions (from the Work Plan) · the live work-phase state of each anchor · the operator's day list.

Outputs. On-site identification and navigation, the operator knows exactly which anchor they are at, and a tap selects it for capture. It produces no records of its own.

KEY INTERACTIONS

→ Read the located map at a glance.

→ Filter by zone.

→ Tap-pin-select into the drill log.

SUCCESS CRITERIA

→ An operator identifies the exact anchor in front of them in seconds, with no ambiguity between adjacent IDs (Brief Tier-1).

→ Every plotted anchor's phase is legible on the map without opening it.

→ An anchor missing a plotted position is visibly flagged, never silently absent.

SCREEN-SPECIFIC RULES

→ Two status surfaces, one truth: the Spatial Map and the Status Board read the same work-phase state; the map locates it on the image, the board tallies it by zone. Neither is authoritative.

→ Viewable by both: the operator uses it on the ground to identify a physical anchor; the PM uses it for located oversight. Same surface, two jobs.

→ Manual plot in V2.0: anchors are plotted by hand on an uploaded image, AI-assisted from the engineer drawing where imported. Full georeferenced or LiDAR placement remains the Site Map roadmap item.

Traceability. `SPM-1 · SPM-2 · SPM-3 · SPM-4 · SPM-5`

PDS-QAQ Drill Summary · QA Queue

Surface. PM · Review & confirm. **Primary user.** Project Manager.

User question. "Which submitted work needs my review and confirmation?"

Purpose. The act layer, where the PM works through submitted anchors, checks them against the evidence requirements, and confirms or reopens. The two-way channel that turns submitted work into confirmed record.

ENTRY POINTS

- → From the Project Home hub (the queue is a top attention item), or by descending from an anchor in review on the Status Board.

INFORMATION HIERARCHY

Primary. The queue of submitted anchors awaiting review, what needs my eyes now, and the act of confirming or reopening. **Secondary.** For the selected anchor, its record and evidence against the requirements: is it complete? **Tertiary.** Filters (by zone, by submitter) and review history.

USER JOURNEY

- → PM opens the QA queue from the hub or board.
- → Picks the next submitted anchor.
- → Reviews its record against the evidence requirements.
- → Confirms it, or reopens it (hands it back) with a reason.
- → Moves to the next.

Inputs. Submitted anchors (from the field) · the Evidence & QA requirements (from setup) · each record's content.

Outputs. Confirmed anchor records (now releasable) · reopened anchors handed back to the field.

KEY INTERACTIONS

- → Review an anchor against requirements.
- → Confirm it.
- → Reopen. Hand back with a reason.

SUCCESS CRITERIA

- → A PM clears the queue efficiently, sees what needs eyes, judges completeness fast, acts.
- → Reopen is one mechanism (the universal hand-back), not several.
- → Nothing is confirmed under-evidenced.

SCREEN-SPECIFIC RULES

- → The act layer: it resolves what the Status Board surfaces.
- → Reopen is the universal hand-back: confirmed or submitted-early, one mechanism.
- → Confirmation is the gate to release: only confirmed records reach Reporting and the engineer (confirmed-only-outward).

Traceability. `DSM-1` · `DSM-2` · `DSM-3` · `DSM-4` · `DSM-5` · `QA-1` · `QA-2` · `QA-3`

PDS-RPT Reporting

Surface. PM · Release. **Primary user.** Project Manager.

User question. "What confirmed work am I releasing to the client, and how?"

Purpose. The release layer, where the PM assembles and releases confirmed work to the engineer as the reporting deliverable. The PM curates what reaches the client; only confirmed records are eligible.

ENTRY POINTS

- → From the Project Home hub or as a project tab.
- → Available once confirmed records exist.

INFORMATION HIERARCHY

Primary. What is being released, the confirmed work selected for the report, and the release action. **Secondary.** The curation controls, what is included or withheld, and by whose discretion (curated by the source of an issue). **Tertiary.** Prior reports and release history.

USER JOURNEY

- → PM opens Reporting.
- → Sees the confirmed work eligible for release.
- → Curates what reaches the client, ground reality tends to the client; contractor performance is resolved first.
- → Generates and releases the report to the engineer.

Inputs. Confirmed anchor records only · the evidence attached · prior reports.

Outputs. A released engineer report, the client-facing deliverable · the record of what was released and when.

KEY INTERACTIONS

- → Select confirmed work.
- → Curate inclusion by source.
- → Generate & release the report.

SUCCESS CRITERIA

- → The PM releases clean, trusted reporting with no chasing, the engineer trusts it on sight (Brief DoD 5, PM to Engineer).
- → Only confirmed work can be released.
- → Curation is for clarity and fairness, not concealment, and is attributable.

SCREEN-SPECIFIC RULES

→ The release layer: only confirmed records are eligible, confirmed-only-outward is enforced here.

→ Curate exposure by the source of an issue: for clarity and fairness, not concealment; edits to released data carry a visible trail.

Traceability. `RPT-1 · RPT-2 · RPT-3 · ENG-1`

PDS-CLO Project Closeout

Surface. PM · Project lifecycle. **Primary user.** Project Manager.

User question. "Is this project complete and ready to close?"

Purpose. The end-of-project wrap, confirming all work is complete, confirmed and released, then closing the project. The terminal state transition for the project as a whole.

ENTRY POINTS

→ From the Project Home hub, as the project nears completion.

→ A deliberate, late-stage action.

INFORMATION HIERARCHY

Primary. Closeout readiness, what remains (any unconfirmed or unreleased work, open items), and the close action. **Secondary.** The final project summary, totals and what was delivered. **Tertiary.** Archived records and the final report set.

USER JOURNEY

→ PM opens Closeout as the project nears done.

→ Reviews readiness, anything incomplete, unconfirmed or unreleased is flagged.

→ Resolves or acknowledges remaining items.

→ Closes the project; it moves to a closed state.

Inputs. The full project state, completion, confirmation and release status across all anchors.

Outputs. A closed project · a final delivered record set · a closed status the org roll-up and the ODC reflect.

KEY INTERACTIONS

→ Review readiness.

→ Resolve / acknowledge remaining items.

→ Close the project.

SUCCESS CRITERIA

→ The PM closes cleanly, confident nothing is unfinished.

→ Anything blocking closeout is obvious.

→ A closed project is unambiguous in status everywhere it appears.

SCREEN-SPECIFIC RULES

→ A deliberate terminal transition: closeout gates on completeness, surfacing anything unconfirmed or unreleased.

→ Closing sets the authored Open / Closed lifecycle flag (PLAT-11), distinct from the derived status; the closed status propagates to the org roll-up and, on the seam, to the ODC.

Traceability. `PLAT-11` · `RPT-1`

PDS-ENG Engineer (Client) View

Surface. External · Read-only. **Primary user.** Engineer (client).

User question. "What confirmed progress can I trust right now?"

Purpose. The commissioning engineer's read-only window into the project, confirmed work only, glance-first, trusted on sight. The external tier of the operator to PM to engineer chain.

ENTRY POINTS

→ The engineer's own login, with their own permissions, scoped to what the PM has released.

→ A separate, external entry, not the PM workspace.

INFORMATION HIERARCHY

Primary. The confirmed state of the job, progress and status they can trust, glance-first. **Secondary.** The released detail per anchor, the evidence and report, one click down. **Tertiary.** Historical reports and prior releases.

USER JOURNEY

→ Engineer logs in to their view.

→ Reads the confirmed state of the job at a glance.

→ Drills into a released anchor's detail if needed.

→ Trusts it, no chasing, no calls.

Inputs. Released, confirmed records only (curated by the PM) · the engineer's permissions.

Outputs. The engineer's trust and awareness, it produces no records. It may acknowledge receipt.

KEY INTERACTIONS

→ Read the confirmed state at a glance.

→ Drill into a released anchor.

→ Acknowledge receipt (optional).

SUCCESS CRITERIA

→ The engineer trusts the reporting on sight, with no chasing (Brief DoD 5, PM to Engineer).

→ The state of the job is immediate; detail is one click away.

→ They never see unconfirmed or withheld work.

SCREEN-SPECIFIC RULES

→ External and read-only: the engineer sees only confirmed, released records, confirmed-only-outward, curated by the PM.

→ The trust surface: its whole job is to be believed on sight; the state of the job must be immediate, detail one click away.

Traceability. `ENG-1` · `ENG-2` · `ENG-3` · `ENG-4`

DESIGN-SYSTEM FOUNDATION · PHASE 3 UX DESIGN ARTEFACT

Design System Spec

The written companion to the live reference: tokens, principles, usage. Updated to Brand Guidelines v1.1.

STATUS	v0.4 · Brand v1.1 aligned	CLASS	Internal
OWNER	Product Owner (founders)	PRODUCT	TACEDGE Ground Engineering (Workspace B1)
COMPANION TO	Design System (live reference)	PHASE	3 · UX Design

· What this is

The written record of the design-system foundation, the source of truth for the tokens, the principles and the usage rules. The live HTML reference shows them in use; this document fixes the values and the discipline.

Every token is a CSS variable that maps directly to Tailwind and shadcn, so the design system and the build share one source of truth. This edition realigns the tokens, naming and type system to Brand Guidelines v1.1 (FA v1.6 · Backlog v0.17 · Brief v3.2 · PDS v1.2).

CHANGES IN THIS EDITION

Aligned to Brand Guidelines v1.1. **Primary Dark #112411** is now the structural primary throughout the UI (Logo Forest #2B4721 is retained for logo and app-icon artwork only). The type system is **Play / Be Vietnam Pro / JetBrains Mono**; the former IBM Plex voice is retired. The radius scale is updated to 8 / 12 / 16.

01 Field-design principles

Five rules decide every component. The product lives outdoors, in gloves, between a remote PM and a client who must trust it on sight, so legibility and trust outrank polish.

→ **Glance-first.** State reads in seconds without opening anything; colour and shape carry status before text does.

→ **Gloved and bright.** A 48px touch floor, high contrast, generous spacing and large type, built for a wet screen in direct sun.

→ **Capture stays sparse.** Operator screens show only what varies; inherited context is read-only and quiet.

→ **Trust is visible.** Freshness, provenance and confirmation are shown, never assumed. Confidently-wrong is worse than absent.

→ **Earthy, not alarmist.** Even alerts stay calm and grounded, a steady instrument, not a dashboard of flashing lights.

02 Colour tokens

BRAND AND STRUCTURE

The v1.1 brand and structure tokens. Primary Dark is primary and structural; Utility Sage is the working mid-tone; the sages and cream are the surfaces.

Primary Dark
#112411 · --primary-dark

Primary Dark Hover
#1A3219 · --primary-hover

Logo Forest
#2B4721 · --logo-forest

Utility Sage
#6E7D5C · --sage

Sage Edge
#AEB89C · --sage-edge

Pale Sage
#B2B594 · --pale-sage

Warm Cream
#F7F5EC · --cream

NAME	HEX · TOKEN	USE
Primary Dark	`#112411 · --primary-dark`	Primary. Structure, headings, the terminal positive states.
Primary Dark Hover	`#1A3219 · --primary-hover`	Pressed and hover of primary.
Logo Forest	`#2B4721 · --logo-forest`	Provisional (D1). Exists only inside logo artwork and the app-icon tile. Never a UI surface colour.

NAME	HEX · TOKEN	USE
Utility Sage	`#6E7D5C · --sage`	Working mid-tone. Active state, secondary buttons, markers.
Sage Edge	`#AEB89C · --sage-edge`	Olive edge, active outline.
Pale Sage	`#B2B594 · --pale-sage`	Quiet surface, planned and in-progress fills, dividers.
Warm Cream	`#F7F5EC · --cream`	Paper, the base background.

PROVISIONAL

Logo Forest #2B4721 is provisional and used only in logo and icon artwork; #112411 Primary Dark is the structural primary throughout the UI.

SURFACES, TEXT AND LINES

NAME	HEX	USE
Card	`#FFFFFF · --card`	Raised surfaces: cards, inputs, the data card.
Sage Tint	`#E6EBD9 · --sage-tint`	Quiet accent panel and information fill.
Quiet Tint	`#EFF1E4 · --quiet-tint`	Neutral fill, one step up from cream, for quiet panels and alternating rows.
Stage	`#DDE3D2 · --stage`	Recessed staging surface behind cards.
Primary Text	`#242A1F · --ink`	Body text and high-contrast labels.
Secondary Text	`#5B6253 · --ink-60`	Supporting copy and secondary labels.
Muted Text	`#646A5A · --ink-40`	Captions, metadata, de-emphasised text.
Divider	`#CBD2C9 · --divider`	Hairline borders and dividers.
Border	`#E7E2D6 · --border`	Card and control borders.
Border Strong	`#D9D3C4 · --border-strong`	Emphasised edges and outlines.

STATUS, THE ONLY NON-BRAND HUES

Beyond the brand palette. Warning and Error are the only non-brand hues, reserved strictly for caution and alert. Never decoration.

NAME	HEX · TOKEN	USE
Warning · caution	`#B07D2B · --warning`	Ageing, submitted, variances. Warning tint #F5EBD4 for fills.
Error · alert	`#9E3B2E · --error`	Issue, test-fail, incident, stale. Error tint #F3DDD6 for fills.

STATE PALETTES

The crux of the system. Two lifecycle dimensions move through states (work-phase and review); two overlays ride alongside them (freshness and the testing marker). A work item carries one value from each at once, so the Status Board and the QA queue can tell the truth simultaneously. Findings sit on top as markers: operator-declared exceptions, and the calmer, system-computed variances.

WORK PHASE · LIFECYCLE, THE HAPPY PATH DARKENS GREEN

STATE	MAPS TO	TOKEN
PLANNED	Pale Sage	`--pale-sage`
ACTIVE	Utility Sage	`--sage`
COMPLETE	Primary Dark, filled	`--primary-dark`
REQUIRES-REDRILL (OFF-PATH)	Error · alert	`--error`
Not-required (off-path)	Muted, de-emphasised	`--divider`

REVIEW · LIFECYCLE, ONLY CONFIRMED IS RELEASABLE

STATE	MAPS TO	TOKEN
IN PROGRESS	Pale Sage	`--pale-sage`
SUBMITTED FOR QA	Warning	`--warning`
CONFIRMED	Primary Dark, filled	`--primary-dark`
RE-OPENED	Error	`--error`

FRESHNESS · OVERLAY, A SECONDARY DOT / RING

STATE	MAPS TO	TOKEN
Fresh	Primary Dark	`--primary-dark`
Ageing	Warning	`--warning`
Stale	Error	`--error`

TESTING · OVERLAY, AN INDEPENDENT BOOLEAN MARKER

STATE	MAPS TO	TOKEN
Flagged for test	Warning · caution	`--warning`
Tested	Primary Dark	`--primary-dark`

EXCEPTION MARKERS · OPERATOR-DECLARED, ALWAYS ALERT-CLASS

STATE	MAPS TO	TOKEN
Redrill	Error	`--error`
Test-fail	Error	`--error`
Incident	Error	`--error`

VARIANCE MARKERS · SYSTEM-COMPUTED, ANNOTATE-CLASS, CALMER

STATE	MAPS TO	TOKEN
Depth variance	Warning · caution	`--warning`
Grout variance	Warning · caution	`--warning`

Off-path work phases sit outside the happy path: Requires-redrill is operator-declared and alerts the PM; Not-required is a phase a work type does not need, shown de-emphasised rather than as a distinct hue. The work-phase fill is always primary; freshness and testing ride as a dot or ring so they never compete with it.

03 Typography

One type system, three roles, self-hosted woff2 (never a font CDN). Play carries display and identity; Be Vietnam Pro carries body and the field UI; JetBrains Mono carries data, so work-item references (e.g. B12), depths and codes read as instrument readouts.

DISPLAY · PLAY 700

Ground anchor B12

Headings and identity. Play (700, also 400). Hero 44 · Section 28.

BODY / UI · BE VIETNAM PRO 400-700

Capture the depth and grout for anchor B12.

Body and labels. Be Vietnam Pro (400 to 700). Body-large 18 (field default) · body 16.

DATA / UTILITY · JETBRAINS MONO 400 / 500 / 700

`B12 · 12.4m · 3 anchors · TOK-8842`

Work-item references (e.g. B12), depths, counts, tokens, codes. JetBrains Mono (400 / 500 / 700). Label 12 · data 15.

Desktop and field share one type system; there is no separate field typeface.

04 Space, radius and elevation

SCALE	VALUES
Spacing	4px base grid · 4 · 8 · 12 · 16 · 20 · 24 · 32 · 40.
Radius	Restrained, instrument feel · sm 8 · base 12 (buttons) · lg 16 (cards) · pill (chips, dots).
Elevation	Low, favour hairline borders over shadow; one soft shadow when raised.

05 Usage rules

→ **Primary Dark is primary and structure.** The work-phase happy path darkens green: Planned (sage) to Active (olive) to Complete (Primary Dark, filled).

→ **Warning and Error are the only non-brand hues.** Reserved strictly for caution and alert states; never used for decoration, emphasis or branding.

→ **The status chip is the workhorse.** One component carries every state dimension (dot, tint, label). The filled Primary Dark variant marks a terminal positive (Complete, Confirmed) and nothing else.

- **Freshness and testing are overlays.** Work-phase is the primary fill; freshness and the testing marker ride as a dot or ring so they never compete with it.
- **Variances are not alerts.** System-computed variances (depth, grout) render in Warning and are the PM's to annotate; only operator-declared exceptions (redrill, test-fail, incident) render alert-class in Error.
- **Confirmed is visually distinct.** Only confirmed work is releasable to the client (confirmed-only-outward), so the confirmed state must always read as terminal-positive.
- **The field is the design target.** A 48px touch floor, high contrast and large type on every interactive element; the field is the target, not the desk.
- **Tokens are CSS variables.** They map one-to-one to Tailwind and shadcn, so the design system and the codebase never drift.

What follows. Next, Batch 2, the ground-engineering-specific components: the work-item pin and its state variants, the Status Board zone grid, the Spatial Map pin (legible on photo and drone backgrounds), the freshness and testing indicators, the variance and exception markers, the JSA sign-on gate, the QA-queue row, an evidence thumbnail with provenance, table and list density, and a dark or high-contrast field mode.

THE LAYER BETWEEN THE FUNCTIONAL ARCHITECTURE AND IMPLEMENTATION

Technical & Data Architecture

How the workspace is shaped beneath the surfaces: data, identity, state, interfaces. The layer between the Functional Architecture and implementation.

STATUS	v0.2 · Draft · §00–§12	CLASS	Internal
OWNER	Product Owner (founders)	PRODUCT	TACEDGE Ground Engineering (Workspace B1)
DERIVES FROM	Backlog v0.17 · FA v1.6 · PDS v1.2 · DS v0.4	PHASE	2 · Architecture
AUDIENCE	Technical Owner · PO · Developers		

How to read this document

This is the architectural-decision and data layer: one level more concrete than the Functional Architecture (how the workspace behaves) and one level less concrete than a technical design (how it is built). It fixes the things that are expensive to change after build: the data model, identity, state, authorization, the work-type configuration mechanism, the evidence model, the emit interface and the required properties of offline capture and sync. It is not a stack specification. The Requirements Backlog remains the source of truth for scope and the Functional Architecture for behaviour; where this document and the backlog differ on scope, the backlog wins. It is organised in three parts: the data spine (§00–§03), the behavioural model (§04–§07), and the interfaces and offline / sync model (§08–§12).

Contents.

Offline capture & sync · 10 ODC emit interface · 11 Coordination-readiness seams · 12 Deferred to the Technical Owner.

Changes in v0.2. The §10 ODC emit is mapped into the five canonical Functional Architecture objects (Project Summary, Status, Progress, Incident, Risk), replacing the document's own object set. Three §02 entities are added: Evidence & QA Requirements, Plant and Project Reference Document. The Project entity carries an authored Open / Closed lifecycle flag, distinct from derived status (PLAT-11). Derivation is re-pinned to Backlog v0.17, FA v1.6, PDS v1.2, DS v0.4, and the §05 cross-reference is updated now that FA v1.6 §07 is aligned. Informed by the React / Vite prototype.

00 Purpose

WHAT THIS DOCUMENT DECIDES

Every prior artefact deferred a set of decisions to "Phase 2" on the grounds that the data model is the one thing that cannot be cheaply changed later. This document makes those decisions. It states what must be true of the system's data and structure for the requirements to hold, in a form a Technical Owner can build against without it being re-derived from five documents.

It deliberately draws a line between what is decided here and what is the Technical Owner's to choose. The architecture below is the former; the technology that realises it is the latter. Holding that line is what lets this document frame the Technical Owner's work rather than pre-empt it.

DECIDED HERE

- The domain data model and its relationships
- Identity and reference scheme (system ID vs human reference)
- The work-type configuration mechanism
- The state model, including the freshness / testing question
- Ownership and the authorization model
- Findings and derived computation (status, variance)
- The evidence and media model
- The ODC emit shape
- Required offline / sync properties and the conflict policy
- Coordination-readiness seams to preserve

DEFERRED TO THE TECHNICAL OWNER

- Database engine and schema dialect

- Application framework and runtime
- Hosting and infrastructure
- The sync engine or library
- API protocol and shape
- Authentication provider
- Concrete identifier format
- Media storage product and delivery
- Performance, scaling and capacity

The full deferred list is restated in §12. It is named explicitly so the omissions read as deliberate framing, not gaps.

01 Architectural position & constraints

This document sits in Phase 2, between the settled Functional Architecture and the implementation a Technical Owner will lead. It is informed by the existing React / Vite prototype: real data shapes and field constraints are already embodied there, and this model is written to match what the prototype proved, not to contradict it.

CONDITIONS THIS MODEL MUST SATISFY

The Functional Architecture (§12) names five capabilities that must remain reachable without being built now. They are the standing constraints on every decision below; each is a boundary the data model must keep open.

CONDITION	WHAT THE DATA MODEL MUST KEEP TRUE
Additional work types	A work type is data, not code. The model must not assume drill / grout / test; a new type that is not a drill log at all must be addable as configuration on the shared spine.
Domain-agnostic skeleton	The capture → QA → confirmed record → curated visibility skeleton stays domain-neutral. Ground-engineering vocabulary (anchor, lithology, grout) lives in a domain layer above the shared model, never inside it.
ODC-expressible	The confirmed record must map cleanly to the five Operational Data Contract objects. Keep it contract-expressible; do not build the contract, which is owned in the Portfolio Architecture.
Coordination read-side	Every project carries a geospatial location; an organisation tier sits above projects; project status is a first-class, queryable property; and a project is addressable from outside itself.

CONDITION	WHAT THE DATA MODEL MUST KEEP TRUE
Multi-contractor interop	A source-neutral identity travels with the confirmed record, so one contractor's data can enter another party's picture without exposing internal or unconfirmed state.

02 Domain data model

The entities below are the spine the rest of this document references. They are described by what they are and what they carry, not by table schema. The model is deliberately abstract at its core (work item, work type, stage record) with ground-engineering specifics (anchor, pile, lithology) appearing as configuration and values, never as structure.

ENTITY CATALOGUE

ENTITY	WHAT IT IS	KEY ATTRIBUTES · TRACE
Setup & configuration		
Organisation	The tier above projects; the account a project belongs to.	Name, projects. Coordination seam (ARCH-2).
Project	The top-level container for everything in a job.	Code, location, declared work types, derived status, owner, authored Open / Closed lifecycle flag. (ARCH-1/3/4, PLAT-7/11)
Zone	A structural division within a project that work is grouped by.	Name, row direction (LTR / RTL), project. (SET-2)
Work Type	A configuration object defining how a kind of work behaves.	Required stages, fields, validations, completion criteria. Anchor and pile in V2.0. (PLAT-2/3/4/5)
Specification	A reusable building block selected into templates and logs.	Drill methods, grout types, grout mixes, lithology options. (SPEC-1–4)
Work Item Template	A reusable template assembled from specifications.	Bar type / spec, hole diameter, max depth, bond length, drill method, grout mix, load. Work type. (DSN-1–3)

ENTITY	WHAT IT IS	KEY ATTRIBUTES · TRACE
Evidence & QA Requirements	A PM-authored rule set: the evidence a record must carry to be confirmable.	Required evidence per stage and work type, applies-to scope, enforced by the QA queue. (Evidence & QA setup)
Plant	A project's plant and equipment library, defined in setup.	Name, type, preset working-hour basis; project. Supervisor may add on site. (DAS-5)
The work		
Work Item	A located, installable element; the unit of work. Anchor and pile are types of it.	Identity, human reference, location, image position, work type, template, zone, status. (PLAT-1/9, SPM-5)
Stage Record	An independent captured record for one stage of a work item.	Captured values, observations, capturer, timestamps, review state. Stages set by work type. (DRL / GRT / TST, PLAT-4)
Finding	A variance or an exception against the planned record.	Type (variance / exception), source, severity, alert, resolution, release. (PLAT-8, ALT, INC)
Evidence	A captured media artefact with provenance, attachable to any parent.	File, kind, capturer, time, source (camera / upload), parent. (PLAT-10)
Field & safety		
Incident / Near-miss	A standalone safety record, internal-only.	Type, injury, severity, media, project, alert. (INC-1–3)
Daily Activity	A resourcing sheet for payroll and invoicing.	Crew and plant hours, auto-totals, edit trail, day, project. (DAS)
Safety Document	An uploaded JSA / SWMS, versioned and acknowledged.	File, version, project, acknowledgements. (SAF, SET-3)
Sign-on	An attributable daily JSA acknowledgement; gates drilling.	Person, project, day, cleared state. (SOS)
Spatial Base Image	An uploaded site image work items are plotted on.	Image, project / zone, dimensions. Work items carry positions on it. (SPM-1)

ENTITY	WHAT IT IS	KEY ATTRIBUTES · TRACE
Project Reference Document	An uploaded reference file a job runs on; a folder, not a DMS.	File, category, project; reuses the evidence / file model; offline, read-only. Distinct from Safety Document. (DOC-1–3)
People & derived		
Person	An operator, supervisor, PM or engineer with an authenticated session.	Identity, role(s), sessions. (User model)
Derived Project Status	A computed rollup of a project's work items, stored for query.	Phase tallies, percent complete, open findings, QA backlog, last activity, coarse state. (PLAT-7)

RELATIONSHIPS, THE SPINE

The cardinalities that hold the model together. These are the structural commitments; the rest is attributes.

RELATIONSHIP	CARDINALITY	NOTE
Organisation → Project	1 – many	The coordination tier above projects.
Project → Zone	1 – many	Work is grouped and ordered by zone.
Project → Work Type (declared)	1 – many	A project declares its types; V2.0 declares anchor and pile.
Project → Specification / Template	1 – many	Reusable building blocks, scoped to the project.
Work Item Template → Work Type	many – 1	A template belongs to one work type.
Zone → Work Item	1 – many	Each work item sits in one zone.
Work Item → Work Type / Template	many – 1	A work item has exactly one type and inherits one template.
Work Item → Stage Record	1 – many	Stages are independent records; which exist is set by the work type.
Work Item / Stage → Finding	1 – many	Findings attach to the item or the stage that raised them.

RELATIONSHIP	CARDINALITY	NOTE
Capture entity → Evidence	1 – many	Evidence has one parent; the parent may be a work item, stage, incident or daily sheet.
Project → Incident / Activity / Safety Doc	1 – many	Project-scoped field and safety records.
Person → Sign-on → Project	1 – many – 1	One attributable sign-on per person, per project, per day.
Project → Derived Project Status	1 – 1	Computed from work items, not authored.
Project → Plant / Reference Doc / QA Requirement	1 – many	Project-scoped setup, reference and QA-rule entities.

DATA-MODEL INVARIANTS

The rules that must hold regardless of implementation. If any of these breaks, a downstream requirement breaks with it.

→ **A work item has exactly one work type and one template.** Its stages are determined by the work type's configuration, never hard-coded. This is what makes a new work type a configuration rather than a rebuild.

→ **Stages are independent records.** A stage record belongs to one work item and can be saved progressively; submission is allowed at any stage, so early escalation is possible without closing the item.

→ **Evidence has exactly one parent, of any capture type.** The parent is polymorphic (work item, stage, incident, daily sheet); provenance travels with every artefact so its origin is never assumed.

→ **A finding is exactly one of variance or exception.** Variance is system-computed; exception is operator-declared. An alert is the surfacing of either, not a stored third kind.

→ **Identity is separate from label.** A work item's system identifier never changes; its human reference can be re-labelled. Nothing external keys off the label.

→ **Only confirmed records are emit-eligible.** Review state is the single gate to external visibility and to the ODC; re-opening withdraws an item from both.

→ **Project status is derived, never authored.** It is computed from the work items and stored for query; it is read upward, not set by hand.

03 Identity & references

Identity is called out on its own because it is both load-bearing and easy to get wrong. The coordination read-side, the ODC emit and multi-contractor interoperability all depend on stable references, and a reference scheme is expensive to retrofit. The scheme is decided here; the concrete identifier format is the Technical Owner's.

WORK-ITEM IDENTITY

Each work item carries two distinct identifiers, and they must never be conflated.

→ **A system identifier.** Immutable, opaque and unique platform-wide. It is what the ODC, Coordination and any external party reference. It is assigned once and never reused or changed.

→ **A human reference.** A readable label such as B12, unique within its project or zone, shown to users. It can be re-labelled (a zone renumber, a correction) without touching the system identifier, because nothing external depends on it.

PROJECT & ORGANISATION IDENTITY

The coordination conditions are identity conditions. A project carries a system identifier and a human project code, a geospatial location (ARCH-1), and must be addressable and enterable from outside itself (ARCH-4). It sits beneath an organisation tier (ARCH-2). None of this builds Coordination; it keeps the project referenceable from above when Coordination reads the contract.

EXTERNAL REFERENCES

The ODC emit and the coordination read-side reference work items and projects by system identifier, never by label. Because the contract is source-neutral, that identity is what lets one contractor's confirmed record enter another party's picture (multi-contractor interoperability) without carrying any internal or unconfirmed state across the boundary.

> DEFERRED TO THE TECHNICAL OWNER
>
> The concrete identifier format (UUID, ULID or other), the project addressing and URL scheme, and reference-collision handling at scale. The scheme above constrains those choices; it does not make them.

04 Work-type configuration

A work type is the mechanism that lets one platform serve many kinds of work. It is data, not code: a configuration object a project declares and the runtime reads. The operational spine (Configure, Capture, Confirm) is fixed; the work type only configures the Capture content and what counts as done. Release is the outcome of Confirm, not a stage. The catalogue carries six ground-engineering work types (anchoring, piling and retaining, drilling, shotcrete, rockfall protection, drainage); anchoring and piling are the two configurations proven in V2.0, and the remaining four are authored, not built.

WHAT A WORK TYPE DECLARES

ASPECT	WHAT IT CONFIGURES	EXAMPLE
Required stages	The Capture stages, their order, and which are required for completion.	Anchor: drill, grout, test. Pile: drill.
Stage fields	The data each stage captures, with type and unit, drawn from project specifications where relevant.	Drill: depth (m), lithology (from the lithology spec), observations.
Validations	Rules applied at capture, so the field cannot submit invalid data.	Depth numeric and positive; lithology must be a project lithology option.
Completion criteria	Which stages must be complete for the item to be complete: the type's Definition of Done. Completion is distinct from submission, which the operator may do at any stage for QA review (§05).	Anchor: drill + grout + test. Pile: drill only.
Template fields	Which template attributes the type uses, so the template form matches the work.	Anchor: bar type, bond length, drill angle. Pile: depth, method.
Report format	The shape of the released record for this type.	Per-type report layout (RPT-5).

ANCHOR AND PILE AS CONFIGURATIONS

The two V2.0 types are not two products, and not the whole catalogue either. They are two values of the same configuration object, proving the spine on the discrete-element pattern; the remaining four catalogue types onboard the same way.

ASPECT	ANCHOR	PILE
Required stages	Drill, grout, test	Drill
Completion	Drill + grout + test	Drill only (a valid pile finishes at drilled)

ASPECT	ANCHOR	PILE
Testing	Typically load-tested	Usually not
Template	Bar / bond / angle / grout	Depth / method
Spine, capture, QA, reporting, visibility	Shared, unchanged	Shared, unchanged

RESOLUTION AT RUNTIME

- → **The runtime renders from config, not from type.** Capture screens, validations and the completion check read the work type's configuration; nothing in the engine names drill, grout or test.
- → **A new work type is a new configuration.** Adding drilling, shotcrete, rockfall protection or drainage means authoring a config object, not changing schema or code, provided the config vocabulary stays expressive enough for non-drill-log work, the standing constraint from the Functional Architecture.
- → **The completion check is generic.** An item is complete when the work type's required stages are complete; PLAT-5 generalises Definition-of-Done to the platform, with no per-type code path. Submission is separate: it is the review hand-off (§05), allowed at any stage, not a completion test.

DEFERRED TO THE TECHNICAL OWNER

The config authoring surface (how a type is defined), the serialization of the config object, and the expressiveness ceiling: which non-drill-log shapes the V2.0 vocabulary already expresses, and which need extension. That ceiling is the one thing worth testing against a real second type before it is relied on.

05 State model

Two questions have moved separately through this product from the start: where the physical work is, and whether the record can be trusted. They are two independent state machines. Two further signals ride alongside them. This section fixes all four and settles which is which.

WORK-PHASE, WHERE THE PHYSICAL WORK IS

A generic lifecycle derived from stage completion against the work type's required stages. The concrete phase label (Drilled, Grouted, Tested) is a read of which stages are complete; the platform state is generic.

FROM → TO	TRIGGER	NOTE
Planned → Active	First stage captured	Work has begun on the item.
Active → Active	Each further stage captured	The concrete phase advances (Drilled → Grouted → Tested).
Active → Complete	All required stages complete	Per the type's configuration; a valid pile reaches Complete at drilled.
Active → Requires-redrill	Operator raises a redrill exception	Off-path; also fires a PM alert.
Any → set / corrected	PM override	The PM may set or correct an off-path state by hand.

REVIEW, WHO OWNS THE RECORD, AND WHETHER IT IS TRUSTED

The trust state machine. Only Confirmed is releasable, and confirmation is the one transition that changes what the outside world can see.

FROM → TO	TRIGGER	NOTE
In progress → Submitted	Operator submits	Ownership passes to the PM; the operator is locked out.
Submitted → Confirmed	PM confirms	The item becomes engineer-visible and emit-eligible.
Confirmed → Re-opened	PM re-opens	Withdrawn from external view until re-confirmed.
Re-opened → Submitted / Confirmed	Re-reviewed	Re-open is the universal hand-back; a who-and-when record is kept.

TWO GATES, AND TWO OVERLAYS

→ **Two gates, and only two.** JSA sign-on gates the start of Capture (drilling); confirmation gates external visibility. Everything else is encouraged sequence, not enforced chain.

→ **Testing-flag, an overlay.** Whether the item is load-tested: a boolean independent of phase and review, set for tested anchors, absent for piles. Keeping it independent is what lets one spine serve types that test and types that do not.

→ **Freshness, an overlay.** How current the displayed status is (fresh, ageing, stale): a recency and trust signal derived from last update and sync state, not a work state. It exists because confidently-wrong is worse than absent; an unrefreshed status must read as ageing, not as fact.

RESOLVED: THE STATE MODEL

Earlier artefacts disagreed on the third dimension, the Functional Architecture naming a testing-flag and the Design System a freshness ramp. Both are real; neither is a lifecycle dimension. The model is two lifecycle dimensions that move through states (work-phase, review) and two independent overlays that ride alongside them (testing-flag, freshness), with findings as markers on top. The Design System (now v0.4) already adopts this, and the Functional Architecture v1.6 §07 is now aligned to it. This document is the authority on the state model.

DEFERRED TO THE TECHNICAL OWNER

The freshness thresholds (when ageing becomes stale) and the precise sync-driven freshness computation, which depend on the sync model in §09.

06 Ownership & authorization

Ownership and authorization are one model here, because who may edit a record is a function of the record's state, not a static role grant. This is the security model: it decides what each role may see and change, and it is the model the confirmed-only-outward boundary depends on.

ROLES

ROLE	MAY EDIT	SEES
Operator	Own in-progress logs	Own assigned items and the current JSA
Supervisor	Activity and compliance records	The project compliance picture and roster
Project Manager	Any record in QA; the release decision	Everything: all logs, findings, the QA queue
Engineer (client)	Nothing	Confirmed records only

A person may hold several roles (a supervisor often also drills); rights are granted per role, never per person.

OWNERSHIP BY REVIEW STATE

The single-owner rule: exactly one party may edit a record at any review state, so a client deliverable never changes silently.

REVIEW STATE	WHO MAY EDIT	EXTERNAL VISIBILITY
In progress	Operator (save at any stage)	None
Submitted for QA	PM only (operator locked)	None
Confirmed	No one	Engineer-visible, emit-eligible
Re-opened	PM only (reverts to QA)	Withdrawn until re-confirmed

ATTRIBUTION & AUDIT

→ **Acknowledgement is attributable.** Sign-on and acknowledgements are made through the person's own authenticated session, never applied in bulk by a supervisor.

→ **Payment and client records carry a trail.** Where a record drives payment or reaches a client (re-open, re-confirm, activity-sheet edits), a who-changed-what-when trail is kept.

THE BOUNDARY THE MODEL PROTECTS

Nothing reaches the engineer or the ODC until a PM has confirmed it; re-opening withdraws it again. Confirmed-only-outward is the single rule the whole authorization model exists to enforce.

DEFERRED TO THE TECHNICAL OWNER

The authentication provider, the session and identity mechanism, and how permissions are enforced (row-level rules, a policy engine, or otherwise).

07 Findings & derived computation

Findings are how the system records work going other than planned, and a project's status is a finding-aware rollup of its work items. Both are computed where they can be and declared where they cannot: the machine does the arithmetic, the human makes the judgement call.

THE FINDINGS MODEL

FINDING	TYPE	HOW IT ARISES	ALERT
Depth variance	Variance	System-computed: actual depth vs design depth, beyond threshold	Variance flag on Overview
Grout variance	Variance	System-computed: derived volume vs expected, beyond threshold	Variance flag on Overview
Redrill	Exception	Operator-declared (grout loss / failure)	Live PM alert on the item
Test-fail	Exception	Operator-declared at submission	Live PM alert on the item
Incident / near-miss	Exception	Operator-declared, standalone	Immediate project-level alert

An alert is the surfacing of a finding, not a stored third kind. Variances are annotate-class (the PM chooses whether to release); exceptions are alert-class.

VARIANCE COMPUTATION

Variances are computed against the design at capture and submission. Depth variance compares actual to design depth; grout variance compares the derived placed volume, from bag counts and the grout mix, to the expected volume. A variance is raised when it crosses a threshold and recorded but not flagged below it. Release is the PM's call.

DERIVED PROJECT STATUS

Project status is the PLAT-7 rollup. It is computed from the project's work items, refreshed as they change, and stored as a queryable property rather than recomputed on open. It carries phase tallies, percent complete (confirmed of planned), open exception and alert counts, QA-backlog depth, last-activity time, and a single coarse state token (Not started, Active, Blocked, Complete). Like the rest of the rollup, the coarse token is system-derived, never PM-declared: Blocked is raised when the project carries an open blocking exception (an unresolved incident, test-fail or requires-redrill) and clears when those resolve. It is the source of the Overview pulse and the ODC Status object.

> THE COMPUTATION BOUNDARY
>
> The system computes measurable variances and the status rollup; it never declares an exception. Exceptions are human judgement (redrill, test-fail, incident) and are declared explicitly. The machine never invents a problem, so accountability stays clear.

DEFERRED TO THE TECHNICAL OWNER

The variance thresholds and their tuning, and the recompute trigger for derived status (event-driven on write versus scheduled), which interacts with the sync model in §09.

08 Evidence & media model

Evidence was fixed as a first-class entity in §02: a captured artefact (photo, video or file) with provenance, attached to any parent. This section states how it behaves, because media moves differently from records and that difference shapes the sync model in §09.

THE MODEL

→ **One entity, a polymorphic parent.** An evidence item has exactly one parent, which may be a work item, a stage record, an incident or a daily sheet. There is no parent-type explosion; the parent reference carries the type.

→ **Provenance travels with the artefact.** Every item carries who captured it, when, and the source: camera (first-hand) or upload (secondary). Provenance is never inferred after the fact.

CAPTURE AND UPLOAD ARE SEPARATE

A record and its evidence sync on different paths. The record is small structured data; the evidence is a large binary. The record can be complete, submitted, even confirmed while its evidence is still uploading, and the interface shows that difference rather than blocking on it. This separation is what keeps field capture instant on a poor connection.

PROVENANCE AND TRUST

Source is a trust signal, not a label. A camera-captured photo is first-hand; an uploaded file is secondary. The distinction feeds the design system's trust-is-visible rule and the freshness overlay: the system shows where an artefact came from rather than presenting all evidence as equal.

DEFERRED TO THE TECHNICAL OWNER

The media storage product and delivery (CDN), thumbnail generation, maximum sizes and accepted formats. The model fixes what an evidence item is and how it attaches; it does not choose where the bytes live.

09 Offline capture & sync

This is the centerpiece and the largest technical risk in the product. The Functional Architecture states that offline reliability is the entire operator thesis: the operator works in poor-connectivity field conditions, and if capture is not instant and lossless on a bad connection, nothing downstream has value. This section fixes the required properties and the conflict policy; it does not choose the engine.

OFFLINE-FIRST CAPTURE

At capture time the device is the source of truth. Every capture (a stage record, an incident, evidence, a sign-on, a daily sheet) is written locally and is immediately usable; sync is a background reconciliation, never a precondition. Nothing in the operator's flow blocks on connectivity, and no feature degrades to read-only when offline.

TWO PAYLOAD CLASSES

Records and media sync on different paths, because they fail differently.

PAYLOAD	SYNCS AS	NOTE
Records (stage, finding, sign-on, activity)	A queue of structured changes	Small; applied in order within an owner's queue.
Evidence (photo, video, file)	Async binary upload, referenced by the record	Large; a record may be complete while its evidence is still uploading.

THE CONFLICT SURFACE, AND WHY IT IS NARROW

The single-owner-at-every-state rule (§06) does most of the work here: at any moment exactly one party may edit a record, so two parties contending for the same record is largely designed out rather than resolved after the fact. The cases that remain are narrow.

CASE	HOW THE MODEL HANDLES IT
Two operators create items concurrently	No conflict: distinct system identities, creates merge.
An owner edits their own record on two devices	Last-write-wins within the single owner; order preserved per record.
Offline edit to a record whose ownership has moved	Server state wins; the stale edit is surfaced for review, never silently applied.
Offline edit to a confirmed record	Rejected: confirmed records are immutable offline. Re-open is a server-side PM act.
Duplicate submission on retry	De-duplicated by idempotency key.

CONFLICT-RESOLUTION POLICY

Creates merge. An owner's edits to their own record apply last-write-wins, in order. Edits that cross an ownership boundary do not overwrite server state; they surface for human resolution. Confirmed records cannot be edited offline. And no offline capture is ever silently dropped. The mechanism that implements this, a CRDT, an operational-transform layer or a simpler ordered queue, is the Technical Owner's; the policy above is the constraint on that choice.

AUTHORITATIVE STATE & REQUIRED PROPERTIES

The server is authoritative for shared and confirmed state and for ownership; the device is authoritative for its own un-synced captures until they land. Sync is bidirectional but asymmetric: captures push up, while ownership, confirmation and derived status pull down. Five properties must hold regardless of engine.

→ **Durable local store.** Captures survive app restart and device sleep before sync; nothing load-bearing lives only in memory.

→ **Idempotent, resumable sync.** A sync interrupted mid-flight resumes without duplicating or losing records; submissions carry idempotency keys.

→ **Partial sync is normal.** Records and evidence sync independently; a record may land before its evidence, and the interface shows the difference.

→ **Ordering within an owner's queue.** A record's edits apply in the order they were made.

→ **No silent loss.** A capture that cannot be applied is preserved and surfaced, never discarded.

FRESHNESS IS COMPUTED FROM SYNC STATE

This is where the freshness overlay (§05) is grounded. A record not yet synced, or a status the field has not refreshed, reads as ageing or stale rather than as fact. The freshness thresholds deferred in §05 are a function of last-sync time, decided alongside the sync mechanism.

DEFERRED TO THE TECHNICAL OWNER

The sync engine and library, the local store technology, the transport, retry and backoff behaviour, and encryption at rest. The required properties and the conflict policy above constrain these; they do not make them. This is the section to revisit first with the Technical Owner in the room.

10 ODC emit interface

The Operational Data Contract is how a confirmed record leaves the workspace for the Coordination layer and other parties. This section fixes the emit boundary and the mapping; it does not define the contract, which is owned in the Portfolio Architecture and remains at v0.1, assumed.

THE EMIT BOUNDARY

Only confirmed records are emit-eligible (§06). Re-opening an item withdraws it from the emit. The contract is source-neutral: it carries system identity, not internal or unconfirmed state, which is what lets one contractor's confirmed record enter another party's picture without leaking the rest.

WHAT MAPS TO THE CONTRACT

The confirmed record maps cleanly to the five contract objects defined consumer-side in the Functional Architecture (§11): Project Summary, Status, Progress, Incident and Risk. This document maps the workspace data into those five; it does not define or re-name them. System identity is the source-neutral reference every object carries, not an object of its own; the granular confirmed work record is reached by descending into the addressable project (ARCH-4), not flattened into the contract; and evidence travels as provenance-bearing references inside the objects, never as binaries.

OBJECT	EMITTED FROM	NOTE
Project Summary	Project system identity and code, location, declared work types	Source-neutral; the reference others key off.
Status	Derived project status (§07)	The queryable rollup, confirmed state only.
Progress	Phase tallies and percent complete, confirmed of planned	Advancement, distinct from the coarse Status token.
Incident	Confirmed incidents and near-misses: type, injury, severity	Project-level; evidence rides as references, not binaries.
Risk	Released variances and exceptions	Only what the PM has released.

DEFERRED AND OWNED ELSEWHERE

The contract's transport, versioning and authoritative object definitions are owned in the Portfolio Architecture, not here. The one risk this carries into Phase 2 is that the contract is still assumed rather than exercised; it is validated only when a second workspace or a real consumer reads it.

11 Coordination-readiness seams

Four conditions from the Functional Architecture (§12) must stay reachable without being built. They are seams, not features: the data model keeps them open so the Coordination layer can attach later, and builds none of it now.

SEAM	CONDITION	WHAT KEEPING IT OPEN MEANS
Geolocation	Every project carries a location (ARCH-1)	Stored from day one, even though V2.0 does not use it.
Organisation tier	An organisation sits above projects (ARCH-2)	A project belongs to an account, not a flat list.
First-class status	Status is queryable, not computed on open (ARCH-3)	Already decided in §07; the seam is that it is exposed upward.
Addressable project	A project is enterable from outside itself (ARCH-4)	A stable reference and entry point, so Coordination can read it.

The ODC (§10) is the seam between the two products: the Operational Workspace and the Coordination layer, joined by one contract. None of the above builds Coordination; together they keep it reachable.

12 Deferred to the Technical Owner

The consolidated list, gathered from every section. It is named in full so the omissions read as deliberate framing. Each choice is the Technical Owner's; the architecture above constrains it but does not make it.

DECISION	CONSTRAINED BY
Database engine & schema dialect	The entity model and invariants (§02)
Application framework & runtime	Free choice; informed by the React / Vite prototype
Hosting & infrastructure	Free choice
Sync engine, local store, transport	The conflict policy and required properties (§09)
API protocol & shape	The emit boundary and entity model (§10, §02)
Auth provider & session mechanism	The ownership and authorization model (§06)

DECISION	CONSTRAINED BY
Concrete identifier format	The dual-identity scheme (§03)
Media storage & delivery (CDN)	The evidence model and provenance (§08)
Variance thresholds & recompute trigger	The findings and derived-computation model (§07)
Freshness thresholds	The state model and sync state (§05, §09)
ODC transport & versioning	The emit shape (§10); contract owned in the Portfolio Architecture
Performance, scaling, encryption at rest	Free choice; the model imposes no obstacle

Completes the Technical & Data Architecture, v0.2. Across its twelve sections it fixes the data, identity, behaviour and interfaces a Technical Owner builds against, and names, here, exactly what is theirs to choose. The architecture is decided; the stack is open. The one assumption still carrying risk into Phase 2 is the ODC: assumed, not yet exercised. The first thing to do with the Technical Owner in the room is §09.